

Maximal Matchings of Complete Tripartite Graphs

OEIS A297487 and the triangle boundary: all fixed order asymptotics,
unmatched vertices, Poisson and Gaussian laws, and inverse thresholds

Two research reports of one external research session
(bundle Reports 176 and 178), merged into one ProveIt report

Both Parts: 3 October 2026 merged: 6 October 2026

Abstract

Let A_n be the number of maximal matchings of the complete tripartite graph $K_{n,n,n}$ with labelled vertices (OEIS A297487, with $A_0 = 1$). This report prints two manuscripts of one research session. Part I proves an expansion of the marked counts to every fixed algebraic order, uniformly for the marking variable in compact subsets of $(0, \infty)$, with a finite coefficient algorithm and four explicit correction polynomials; its leading term is Václav Kotěšovec's OEIS equivalent $A_n \sim 3\sqrt{2}(2n/e)^{3n/2}e^{\sqrt{n/2}-3/8}$, credited as such. It shows that the number of unmatched vertices of a uniformly chosen maximal matching is within $O(n^{-1/4})$ in total variation of a parity-conditioned Poisson law of mean $\sqrt{n/2}$, with refined mean and variance, a central limit theorem and a local law; it expands the exponentially small probability that a uniform maximal matching is maximum to every fixed relative order; and it brackets the threshold $\min\{n : A_n \geq y\}$ between two ceilings, with a Lambert- W approximation of the smooth inverse. Part II passes to unequal parts. For compact sets of strictly triangular part proportions it compares the three unmatched-part branches to leading order, with a softmax law for the unmatched part near balance. For $K_{2n+d,n,n}$, across the boundary of the triangle inequality, it proves an all-fixed-orders expansion in powers of $n^{-1/3}$, uniform while $d/n^{2/3}$ stays in a compact interval, with exponential suppression of the other branches, Gaussian laws for the unmatched count on the scale $n^{1/3}$, four explicit corrections for the boundary counts $M(2n, n, n)$ (a sequence with no OEIS entry), a two-ceiling threshold bracket, and, for d/n in a compact subset of $(0, \infty)$, an all-orders Poisson expansion. Part II settles one instance of the unequal-parts half of Part I's last question; the case of more parts, uniformity between the regimes and unequal small parts near a degenerating triangle remain open. The write checks the 100 b-file terms of A297487 against Part I's exact counts, recomputes both printed prefixes from the generating function, places the inverses relative to the repository's transseries volume, and collects the open questions. All error constants and onsets are existential. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit f7e9e5c2f (batch 109, cluster 109-TRIP). Both are dated 3 October 2026, and both count maximal matchings of complete tripartite graphs with labelled vertices and distinguished parts, a matching being *maximal* if no edge can be added and *maximum* if it has the largest possible number of edges. The common starting point is that a matching of a complete multipartite graph is maximal exactly when its unmatched vertices lie in one part (Part I, Lemma 2.1; Part II, Lemma 13.1), so that the counts are sums of explicit factorial ratios over the unmatched part and the number of unmatched vertices, with the exponential generating function $(e^{vx} + e^{vy} + e^{vz} - 2)e^{xy+xz+yz}$ (Part I, (6); Part II, (58)).

Part	Bundle report	Delivered title	Labels
I	176 (base)	Maximal Matchings in Balanced Complete Tripartite Graphs: All fixed order asymptotics, unmatched vertices, and inverse thresholds	tmm:bal:
II	178	Maximal Matchings Across the Triangle Boundary: Boundary asymptotics, a uniform moving window, and an exterior Poisson law	tmm:bd:

Arrangement. The base of the merge is Report 176; its `report.tex` was staged as this `article.tex`, and it is printed first, as Part I. It is the report on A297487 itself, the origin of the definitions both Parts use, and the predecessor: Report 178 says that it continues “the balanced $K_{n,n,n}$ analysis of Report 176”, cites it, and takes from it the marked generating function, the maximality lemma and the gamma–Poisson method. The two do not prove the same theorems. Part II’s strict-triangle Theorem 14.1 contains the balanced case only to leading order, with relative error $O(n^{-1/2})$, where Part I is all-order; Part II’s other results concern $K_{2n+d,n,n}$, which is balanced only when $d = -n$, outside all its asymptotic regimes. Report 178 is printed second, as Part II. Part II’s Section 12 recalls that Part I leaves unequal parts and feasibility boundaries as further questions; the section “Part II and the question of Part I” below says exactly what it settles.

Numbering. Sections are numbered continuously, statements within sections, and equations continuously through the report, as in both manuscripts. Part I keeps Report 176’s Sections 1–11 and all its statement and equation numbers. Report 178’s Section k is Section $k + 11$ here (Sections 12–25), its statements move with their sections (its Theorem 4.1 is Theorem 15.1), and its equation (k) is equation $(k + 57)$ (its (1) is (58)); the README maps them. Section 26 was added in the write.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (6 October 2026). They give cross-references between the Parts and record what the write checked. No delivered statement was changed and no delivered symbol was renamed. Section 26 gathers the open questions of both Parts under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing.

What is proved, and where

Statement	Where
<i>Exact enumeration (both Parts)</i>	
Maximal $\iff$ unmatched vertices in at most one part; marked EGF $(e^{vx} + e^{vy} + e^{vz} - 2)e^{xy+xz+yz}$ (classical, elementary)	Part I, Lemma 2.1, (6); again Part II, Lemma 13.1, (58)
Balanced counts $t(n, k)$ and $A_n(v) = 3S_n(v) - 2[2 \mid n]B_n$ (Howroyd's OEIS program is the case $v = 1$)	Part I, (3), (5)
Branch terms for K_{p_1, p_2, p_3} ; the large-part branch; the boundary formula $M(2n, n, n) = (2n)! Z_n$, $Z_n = {}_2F_2(-n, -n; \frac{1}{2}, 1; \frac{1}{4})$, a shifted Hermite moment (classical consequences, not claimed new)	Part II, (59), (60), (64), (65)
<i>Balanced graphs $K_{n, n, n}$ (A297487)</i>	
Global bound $R_n(k) \leq e^2$; Poisson representation	Part I, Lemma 3.1
Marked expansion to every fixed order, uniform for real v in compacts of $(0, \infty)$; $q_1, \dots, q_4$	Part I, Theorem 4.1, (21)–(24)
$A_n = \frac{3}{2}B_n e^{\lambda-3/8}(Q_J(\lambda) + O(\lambda^{-J-1}))$; four corrections to Kotěšovec's equivalent; $\log A_n$	Part I, Corollary 5.1; leading order also Part II, Theorem 14.1 at $p_i = n$
Mean and variance of the unmatched count U_n	Part I, Theorem 6.1
d_{TV} to a parity-conditioned Poisson law $O(n^{-1/4})$; CLT; span-two local law	Part I, Theorem 7.1; CLT also Part II, Corollary 14.2 at $\delta_i = 0$
Probability that a uniform maximal matching is maximum, to every fixed relative order	Part I, Theorem 8.1
Monotonicity; two-ceiling bracket for the threshold; Lambert- W approximation of the smooth inverse	Part I, Lemma 9.1, Theorems 9.2 and 9.3
<i>Unequal parts</i>	
Compact strictly triangular proportions: branch sums to leading order (relative error $O(n^{-1/2})$), conditioned Poisson laws per branch; softmax law for the unmatched part near balance	Part II, Theorem 14.1, Corollary 14.2
$K_{2n+d, n, n}$ with $d/n^{2/3}$ in a compact interval: expansion to every fixed order in $n^{-1/3}$, rational coefficients $c_r(u)$, suppression of the other branches	Part II, Theorem 15.1, Sections 16–18
Window moments, local, total-variation and Kolmogorov bounds $O(n^{-1/3})$	Part II, Theorem 19.1
Boundary $M(2n, n, n)$: $c_1, \dots, c_4 \in \mathbb{Q}(4^{-1/3})$, log, moments, marked check, maximum matchings as an exact rare event	Part II, Corollary 20.1, Section 20
$M(2n+2, n+1, n+1) \geq 2M(2n, n, n)$; two-ceiling bracket; Lambert initializer	Part II, Lemma 21.1, Theorem 21.2
d/n in a compact subset of $(0, \infty)$: all-orders Poisson expansion; the edge deficit tends to Poisson(e^{-2})	Part II, Theorem 22.1
<i>Open</i>	
Exponentially accurate parity sector; markings tending to 0 or ∞ ; complex marking; effective constants and certified thresholds (both Parts); more than three parts; uniform transitions between the regimes; unequal small parts near a degenerating triangle; higher orders in the strict triangle; sharper tails; the boundary sequence's identity and the unread literature. See Section 26.	

The finite computations of the two packages (exact integers and rationals, and exact arithmetic in $\mathbb{Q}(4^{-1/3})$) check identities, coefficients, transcriptions and small counts; no asymptotic statement rests on them, and the optional high-precision diagnostics of Part II are not certificates.

The OEIS entry, and the boundary sequence

The write read **A297487** [7] on 6 October 2026 (revision #18 of 6 February 2026). Its name is “Number of maximal matchings in the complete tripartite graph $K_{\{n,n,n\}}$.” The author is Eric W. Weisstein (30 December 2017); the terms from $a(6)$ on (30 December 2017), the PARI program (dated 30 December 2017) and the b-file “Table of n , $a(n)$ for $n = 1..100$ ” are Andrew Howroyd’s; the offset is 1, so the entry starts at $A_1 = 3$ and the Parts add $A_0 = 1$. On 6 February 2026 Václav Kotěšovec added an order-five recurrence with polynomial coefficients and the equivalent

$$a(n) \sim 3 * 2^{((3*n+1)/2)} * \exp(\sqrt{n/2} - 3*n/2 - 3/8) * n^{(3*n/2)},$$

which is Part I’s (1), since $2^{(3n+1)/2} n^{3n/2} e^{-3n/2} = \sqrt{2} (2n/e)^{3n/2}$. The entry’s Mathematica line, which carries no name, is the “hypergeometric expression” of Part I’s Section 1:

$$3 \, n! \, \text{HypergeometricPFQ}[\{(1 - n)/2, -n, -n/2\}, \{1\}, -4] - \text{If}[\text{Mod}[n, 2] \\ == 0, 2 \, (n!/(n/2)!)^3, 0].$$

Howroyd’s program computes

$$[2 \mid n] \left(\binom{n}{n/2} (n/2)! \right)^3 + \sum_{0 \leq k \leq (n-1)/2} 3 \binom{n}{k}^2 \binom{n}{2k} \binom{2k}{k} k!^2 (n-k)!.$$

[write] This is (5) at $v = 1$ term by term: the bracket is $(n!/(n/2)!)^3 = B_n$, and the k -th summand is $3(n!)^3/((n-k)!k!^2(n-2k)!) = 3t(n, n-2k)$ by (3), so k is the number of edges joining the unmatched part to each of the other two parts (Part I’s $a = b = (n-K)/2$ for $K = n-2k$ unmatched vertices), and $0 \leq k \leq (n-1)/2$ is exactly $K \geq 1$. Hence Howroyd’s sum is $[2 \mid n]B_n + 3 \sum_{K \geq 1} t(n, K) = 3S_n(1) - 2[2 \mid n]B_n$, which is Part I’s “positive finite sum” and its Section 2.

The b-file. Part I checked the 16 displayed terms and says that the b-file was not retrieved. The write retrieved it (100 terms) and compared it with the exact counts $n = 0, \dots, 100$ of Part I’s certificate (`data/176-balanced-data-certificates.json`): all 100 terms agree, and Howroyd’s formula, evaluated independently, gives the same numbers.

The boundary sequence. Part II’s $A_n = M(2n, n, n)$, beginning 1, 3, 74, 4506, 489240, 82306920, $\dots$, is *not* A297487 (Part II says so), and Part II found no OEIS entry for it. The write’s OEIS searches of 6 October 2026 for 74,4506,489240,82306920 and for 4506,489240,82306920,19743705360 returned “No results”. So the sequence has no OEIS entry as far as the write can see; that is evidence, not a proof of absence. Nothing was submitted to the OEIS.

Part II and the question of Part I

Part I’s last question (Section 11.2, item 5) reads: “**Unequal parts and more parts.** For $K_{a,b,c}$ or a growing number of parts, identify the feasibility boundaries and competing unmatched-part contributions. Symmetry is central to the simple scalar gamma ratio used here.” For part sizes p_1, p_2, p_3 , Part II’s branch formula (59) shows which unmatched part is possible: with k vertices unmatched in part i the three edge populations are $a_i + k/2$, $a_j - k/2$, $a_\ell - k/2$, $a_i = (p_j + p_\ell - p_i)/2$, so a branch other than i is feasible only if $a_i \geq 0$. When $p_i > p_j + p_\ell$ only branch i survives; inside the triangle all three compete. The *feasibility boundaries* are therefore the faces $p_i = p_j + p_\ell$ of the triangle inequality.

Part II's object, $K_{2n+d,n,n}$, is literally a $K_{a,b,c}$ (with $a = 2n + d$, $b = c = n$), and $d = 0$ is the face $a = b + c$. Part II therefore *settles one instance of the unequal-parts half* of the question: for this isosceles family it identifies the boundary behaviour to every fixed order in the window $d/n^{2/3}$ compact (Theorem 15.1), proves that the competing branches are exponentially suppressed there ((75)), and gives the exterior law for d/n compact in $(0, \infty)$ (Theorem 22.1). It also *advances* the general three-part case: for any compact set of strictly triangular proportions, Theorem 14.1 identifies the competing contributions to leading order (relative error $O(n^{-1/2})$, not all orders), and Corollary 14.2 gives the softmax law for the unmatched part near balance. Open, and recorded in Section 26: the *more parts* half (item 5); faces with unequal small parts, $p_1 = p_2 + p_3$ with $p_2 \neq p_3$, and the regime where several branches stay comparable near a degenerating triangle (item 7, Part II's own question 4); a theorem uniform as $d/n^{2/3} \rightarrow \pm\infty$ joining the three regimes (item 6); and higher orders in the strict triangle.

The inverses and the transseries volume

The repository's transseries volume [11] has a Part 0 apparatus for inverting rapidly growing functions. Its symbols are subscripted “vol” here where they meet this report's letters (κ_{vol} , d_{vol} , v_{vol} , X_{vol} , L_{vol} , Φ_{vol} , ϱ_{vol}): Part II's d is the imbalance and its κ_r are coefficients, and v is a marking variable in Part I and a variance in Part II. Statement by statement **[write]**:

- Part I, the initializer of Theorem 9.3: $x_0 = 2\ell/(3w)$ with $w = W(4\ell/(3e))$ and $\ell = \log y$, the root of $H(x) = (3x/2)\log(2x/e) = \ell$. An *instance* of the factorial core **p0:prop:factorial-core** with $\kappa_{\text{vol}} = \frac{3}{2}$, $d_{\text{vol}} = \frac{3}{2}(\log 2 - 1)$ and $L_{\text{vol}} = \ell > 0$, since $H(X) = X(\kappa_{\text{vol}} \log X + d_{\text{vol}})$. The proposition gives $v_{\text{vol}} = W_0((2\ell/3)e^{\log 2 - 1}) = W_0(4\ell/(3e)) = w$ and $X_{\text{vol}} = e^{v_{\text{vol}} - d_{\text{vol}}/\kappa_{\text{vol}}} = e^{w+1}/2$; as $w e^w = 4\ell/(3e)$, $e^{w+1}/2 = 2\ell/(3w) = x_0$. Its side condition is $\kappa_{\text{vol}} \log X_{\text{vol}} + d_{\text{vol}} = \frac{3}{2}w > 0$, and its slope formula gives $H'(x_0) = \kappa_{\text{vol}}(v_{\text{vol}} + 1) = \frac{3}{2}(w + 1)$, which is Part I's D .
- Part I, the corrections of (56). Part I's proof inverts $H + g$, $g(x) = \sqrt{x/2} + C$, by Taylor expansion and finds $\delta = -g/D + g'g/D^2 - H''g^2/(2D^3) + \dots$. These three terms are exactly the terms $m = 1$ and $m = 2$ of the perturbation-operator series **p0:eq:operator-series** (**p0:prop:operator-form**) with $\Phi_{\text{vol}} = H$, $\varrho_{\text{vol}} = g$, $\varepsilon = 1$ and $X = x_0$: the term $m = 1$ is $-g/H'$, and the term $m = 2$ is $\frac{1}{2}(H')^{-1} \frac{d}{dX} [g^2/H'] = gg'/(H')^2 - g^2 H''/(2(H')^3)$. The proposition is a formal series; the error $O_J(1/(\sqrt{x_0}D))$ and the localization are Part I's own. An *analogue*, not an instance of a theorem of the volume.
- Part II, the initializer of Section 21: $x_0 = \log y/(2W(\log y/e))$, the root of $2x \log(2x/e) = \log y$. An *instance* of **p0:prop:factorial-core** with $\kappa_{\text{vol}} = 2$, $d_{\text{vol}} = 2(\log 2 - 1)$, $L_{\text{vol}} = \log y$: $v_{\text{vol}} = W_0(\log y/e)$, $X_{\text{vol}} = e^{v_{\text{vol}}+1}/2 = \log y/(2v_{\text{vol}}) = x_0$, and the slope $\kappa_{\text{vol}}(v_{\text{vol}} + 1) = 2\log(2x_0)$ is the “leading derivative” Part II names.
- Part II, the leading displacement $x_M - x_0 \sim -3ax_0^{2/3}/(2\log(2x_0))$: the term $m = 1$ of **p0:eq:operator-series** about the same core with $\varrho_{\text{vol}}(x) = 3ax^{2/3}$, as an asymptotic statement proved in Part II. An *analogue*. Part II says that it is “only a leading displacement, not a complete ordered reversion series”.
- Part I's Theorem 9.2 and Part II's Theorem 21.2, the two-ceiling brackets: each is proved directly at the integers, from the monotonicity of the sequence, a smooth model (F_J , resp. B_M) and the relative error at integer arguments, without an interpolation of the sequence. They are *not* instances of **p0:thm:staircase**. When the two ceilings agree they determine

the threshold, the conclusion of item (2) of that theorem; both Parts' warnings that a bare ceiling of the smooth root is not proved describe the failure mode of its items (2) and (5).

Neither Part claims novelty for the inversion mechanics.

Notation across the two Parts

Common to both Parts with the same meaning: U (the number of unmatched vertices of a uniform maximal matching; Part I writes U_n), the macros $\mathbb{E}$, $\mathbb{P}$, Var , Poisson, d_{TV} , $(x)_r$, $\psi = (\log \Gamma)'$, B_{2m} (Bernoulli numbers), W (the real Lambert branch), “maximal” and “maximum”. Each Part fixes its own meaning of the symbols below and opens with a short table of its readings. No symbol was renamed.

Symbol	Meanings
A_n	Part I: the balanced count $M(n, n, n)$, A297487. Part II: the boundary count $M(2n, n, n) = A(n, 0)$. <i>Tempting false reading:</i> Part II's A_n is not A297487, and Part II's Lemma 21.1 is not about Part I's sequence.
M	Part II: $M(p_1, p_2, p_3)$ the count, $M_{p_1, p_2, p_3}(w)$ the marked count, and M the truncation order. Part I: $M_{a, b, c}(v)$ the marked count; M a truncation order in (35) and a window constant in Theorem 7.1.
λ	Part I: $\sqrt{n/2}$. Part II: $\lambda_i = \sqrt{a_j a_\ell / a_i}$ the branch means (all equal to Part I's λ at balance), $\lambda = \sqrt{a_1}$ in Section 16, and $\lambda = c^{-2}$ in Section 22.
J	Part I: a truncation order. Part II: the random number of edges between the two parts of size n . <i>False reading:</i> Part II's J is a random variable, not an order.
t, k, j	Part I: $t(n, k)$, k the number of unmatched vertices. Part II: $t(n, d, j)$, j the number of edges between the two small parts (at $d = 0$, $k = 2j$); k unmatched vertices in the branch terms $T_i(k)$, and an index in c_k .
v, w, z	Part I: v the marking variable; $z = k/\lambda$. Part II: w the marking variable of (58) (Part I's v), $v = \chi^{-1}$ the Gaussian variance, z an EGF variable and the marking variable of Section 20. The volume's v is v_{vol} .
q	Part I: the polynomials $q_j(v)$, Q_J . Part II: $q = N^{-1}$.
a, b, c	Part I: $a_j = q_j(1)$; the marking compact $K = [a, b]$; a, b, c edge numbers in Section 2; $a = (J + 1)/2$ in a proof; c_K, c_J constants. Part II: $a = 4^{-1/3}$, the edge populations a_i ; $b = -2s/\chi$ the Gaussian centre; $c = d/n$ in Section 22; $c_r(u)$, c_k the coefficients; c, c_K constants.
B	Part I: $B_n = (n!)^3/\Gamma(n/2 + 1)^3$; $B_{m+1}(\cdot)$ Bernoulli polynomials. Part II: B the gamma reference (equal to B_n at balance), $B_M(x)$ the smooth boundary model.
d, D	Part I: $d_j(z)$ the Stirling polynomials; $D(h, z) = \log R_n(k)$; $D = \frac{3}{2}(w + 1)$ in Theorem 9.3. Part II: d the imbalance; $D = \delta_1 + \delta_2 + \delta_3$, $D = \max(0, -\inf K)$, $D_{n, d}$, D_M . The volume's d is d_{vol} . <i>False reading:</i> Part I's d_j are not related to Part II's d .
h	Part I: $h = \lambda^{-1}$. Part II: the phase polynomials $h_r(y)$, and a summation index in (77).
G, H	Part I: $G(X)$ the formal Stirling expression, G_n the gap from maximum size; $H(x) = (3x/2) \log(2x/e)$. Part II: $G(q, y)$ the formal phase; $H(n, d)$ the window normalizer (73); $H_*(y)$ the threshold.
N, y	Part I: $N(y)$ the threshold; $y > 0$ its argument. Part II: $N = n^{1/3}$; y the local variable $(j - sN^2)/N$, and the threshold argument in Section 21.
R, T, S	Part I: $R_n(k)$ the gamma ratio; $T_r(h)$ the inserted series; $S_n(v)$ the branch sum. Part II: $R_i(k)$ the branch ratio (Part I's R_n at balance), R_j the exterior ratio, $R_{M, n, d}$ a remainder; $T_i(k)$ the branch terms, T_0 ; S_i the branch sums (Part I's $S_n(1)$ at balance), $S_m(x)$ Faulhaber polynomials.
C	Part I: $C = \log(3\sqrt{2}) - 3/8$; generic constants. Part II: $C_i = \frac{\lambda_i^2}{8} \sum a_r^{-1}$, the phase constant C of (72), $C(q)$, C_K , C_D , C_M .

Symbol	Meanings
P, Q, E, F	Part I: $P \sim \text{Poisson}(v\lambda)$, $p_j(z)$; $Q_J(v, \lambda)$; $F_J(x)$ the smooth model. Part II: $P(x)$ a polynomial, $P_k(y)$, $P \sim \text{Poisson}(c^{-2})$; $Q_{n,d}$ the lattice Gaussian; E_r , $E_{M,n,d}$; $F = p_1!p_2!p_3!$.
K	Part I: $K = [a, b]$ the marking compact, $K^* = [a/2, 2b]$. Part II: K the compact δ -interval, K_+ the compact c -set.
μ, δ, ℓ	Part I: $\mu_r(t)$ centred Poisson moments; δ the inverse correction; $\ell = \log y$, ℓ_m . Part II: μ_r Gaussian moments; $\delta = d/N^2$, δ_i ; ℓ_k , $\ell_2(u)$ logarithmic coefficients.
x, X	Part I: $x = \lambda^2 = n/2$, real model variable; X the Stirling variable. Part II: x, y, z EGF variables, $x = u^3$ in Section 19, x a real model variable; $X \sim \mathcal{N}(0, 1)$ in (65). The volume's X is X_{vol} .
κ, L, Φ	Part II: $\kappa_r(c)$ exterior coefficients; L the lower summation limit (61); Φ the window phase (72). Part I: L_n the right side of (1). The volume's are κ_{vol} , L_{vol} , Φ_{vol} .
ε, Δ	Part I: $\varepsilon_J(y)$ the bracket half-width; $\epsilon_n = (-1)^n$. Part II: Δ_M the bracket half-width; ε a small exponent.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 3 October 2026. Their author lines read “Research report 176” and “Research report 178”; neither names a person, a tool or an addressee, and neither carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by f7e9e5c2f (batch 109, cluster 109-TRIP).

- **Part I**, bundle Report 176, archive `Tripartite_Maximal_Matchings_Asymptotics_and_Inverses_Source.zip` (550,292 bytes; 20 files; `report.tex`, 628 lines, 14 pages); the base. Contributes the balanced theory, the definitions and the marked generating function. Pin: none named; its `SOURCES.md` cites the collection’s `manifest.tex` and `README.md` by blob (9b13df17, 398c8bff), the versions committed by ca62e1488 (2 October 2026) and replaced by 7969f7168 (3 October 2026, 09:49 Pacific); so the session read the catalogue as of ca62e1488.
- **Part II**, bundle Report 178, archive `Unequal_Tripartite_Matchings_Transitions_and_Inverses_Source.zip` (633,432 bytes; 31 files; `Report178.tex`, 755 lines, 19 pages). Contributes the unequal-parts theory. Pin: none (its audit cites the “then-current” manifest and README).

Where the merge had to choose:

1. *Base and order*. Report 176 first, as the base and the predecessor; 178 second (see the Guide).
2. *Repeated material*. Both Parts prove the maximality lemma, state the marked generating function, and use the same digamma inequalities and Poisson tilting. Part II states these again because its proofs cite them, and its own text says that the generating function is “already recorded in Report 176”. Both are printed as delivered, with **[write]** pointers; nothing was dropped.
3. *Front matter, numbering, layout*. The two title blocks and abstracts are printed at the heads of their Parts; one table of contents replaces two; the two identical macro sets are merged (Part II adds $\mathcal{N}$); the two manuscripts redefine the table-of-contents line for sections differently, and the merge keeps Report 176’s (bold entries); the running head names the Part. Part II’s equations are renumbered by the offset 57 (see the Guide).

4. *Bibliography.* The two bibliographies are merged into one, keeping every detail of every entry; where both Parts cite a work with different annotations or links (`oeis`, `song`, `azor`, `paris`), both are kept and attributed. Both manuscripts use the key `dlmf`, for different sections of the DLMF (Report 176 for §5.11 and §24.2, Report 178 for §16.11): Part II's key is renamed `dlmf16`. Part II's `r176` is Part I; the entry is kept, with a note. The write added the transseries volume.
5. *Write additions.* This front matter, the reading-conventions tables of the Parts, the dated notes, Section 26, labels for Part I's first section and last subsection (`tmm:bal:sec:convention`, `tmm:bal:sec:questions`), the two Part labels, and the label prefixes. Everything else is delivered text. The verification markers of both manuscripts (the display after “The first values” in Part I, the comment-delimited prefix in Part II) are kept unchanged, so that both delivered verifiers still parse this file (README).

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 109, 6 October 2026) both manuscripts were read and no mathematical step was found false. Checked there: literal enumeration of maximal matchings, giving A297487 for $n \leq 4$ (1, 3, 14, 342, 5256), $M(2n, n, n)$ for $n \leq 3$ (1, 3, 74, 4506), and $M(4 + d, 2, 2) = 14, 50, 74, 202, 482, 1010$ for $d = -2, \dots, 3$; at $n = 2000$, 2001 and 20000, λ^4 times the remainder of A_n/L_n after three corrections is 0.1810, 0.1810, 0.1834, against Part I's fourth coefficient $1879441301/10192158720 = 0.1844$; for the boundary at $n = 10^4$ and 10^5 , N times the relative remainder is 0.6219 and 0.6185 against $c_1 = 211a/216 = 0.6154$, and the next scaled residual 0.139 and 0.144 against $c_2 = 172781a^2/466560 = 0.1470$ (rounded); Part I's Stirling terms in $\log B_n$ and Part II's exterior κ_1 by hand; and the delivered suites on copies (README).

The write (6 October 2026) read the OEIS entry and its b-file and compared all 100 terms with Part I's exact counts; evaluated Howroyd's program independently and proved that it is (5) at $v = 1$ (above); recomputed Part I's displayed prefix $A_0, \dots, A_8$, Part II's displayed prefix $M(2n, n, n)$, $n \leq 10$, and the six values $M(4 + d, 2, 2)$ above by exact coefficient extraction from the generating function (6) (all agree); searched the OEIS for the boundary sequence; checked the specializations of Part II's Theorem 14.1 and Corollary 14.2 at balance against Part I (notes in Sections 5 and 7); checked $c_1(2a) = 211a/216$ from (94), the constant $-83/648$ and the term $\frac{1}{6} \log n$ of (107), and the identities of the two preceding sections; ran both delivered verifiers against this merged `article.tex` on copies (README); and built the two delivered manuscripts separately to confirm every delivered label under the prefixes and offsets.

What was *not* done: the analytic proofs (uniform Stirling and Chernoff estimates, Taylor and lattice remainders, the saddle localization, the suppression bound, the transfer to limit laws) were read and their displayed algebra spot-checked, but they are the manuscripts' proofs, not an independent verification. The full texts of Song–Miao–Wang–Zhao, Azor–Gillis–Victor and Luke were not obtained by the Parts or by the write; Paris, Hang–Luo, Drmota et al. and DLMF were not re-read by the write. The PDF and ZIP builders were not run, and the optional mpmath diagnostics of Part II were not rerun.

Relation to neighbouring reports and formal projects

No other placed report of the repository treats A297487, maximal matchings of complete multipartite graphs, or the boundary family $K_{2n+d,n,n}$ (repository search, 6 October 2026). The collection's

`matching-rank-normalization` and `preorder-root-polytopes` use the word “matching” for unrelated objects. The transseries volume [11] is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats matchings of complete multipartite graphs, and no statement of this report is formalized.

Part I

Maximal matchings in balanced complete tripartite graphs: all fixed order asymptotics, unmatched vertices, and inverse thresholds

Bundle Report 176, archive `Tripartite_Maximal_Matchings_Asymptotics_and_Inverses_Source.zip`, delivered as `report.tex` (14 pages), the base of the merge. Delivered title block “Maximal Matchings in Balanced Complete Tripartite Graphs / All fixed order asymptotics, unmatched vertices, and inverse thresholds / Research report 176 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. The labels of Section 1 and Subsection 11.2 are added in the write; the running head “Report 176” became the Part’s head.

Abstract of this Part (as delivered). Let A_n count maximal matchings of the labeled complete tripartite graph $K_{n,n,n}$, the sequence OEIS A297487. We derive a real-positive marked expansion to every fixed algebraic order, uniformly on compact marking intervals, with an explicit finite coefficient algorithm and the first four correction polynomials. Its leading term is the equivalent already recorded by Václav Kotěšovec in OEIS. The positive finite sum underlying the proof is already implicit in Andrew Howroyd’s OEIS program.

For a uniform maximal matching, the number of unmatched vertices is within $O(n^{-1/4})$ in total variation of a parity-conditioned Poisson law of mean $\sqrt{n}/2$. Direct inserted-moment expansions give refined means and variances; a central limit theorem, a span-two local law, and all-fixed-order relative expansions for the rare maximum-matchings probabilities follow. Smooth asymptotic models have inverses with arbitrarily high fixed algebraic accuracy and boundary-safe two-ceiling enclosures for integer thresholds. A Lambert W approximation gives an $o(1)$ -accurate real inverse.

All error constants and onsets are existential. No exponentially accurate count transseries, complex-marking theorem, unconditional ceiling rule, or global priority claim is made. Exact offline verification accompanies the proofs.

Reading conventions of Part I `[write]`

$A_n, A_n(v), S_n(v), t(n, k)$	A297487 with $A_0 = 1$; the marked count; one branch sum; the count with k unmatched vertices (Part II’s A_n is a different sequence)
λ, h, x, B_n	$\sqrt{n}/2, \lambda^{-1}, n/2, (n!)^3/\Gamma(n/2 + 1)^3$ (Part II’s B at balance)
$R_n(k), P, K, K^*$	gamma ratio (Part II’s R_i at balance), Poisson($v\lambda$), marking compacts
d_j, p_j, q_j, a_j, Q_J	Stirling polynomials (unrelated to Part II’s imbalance d), their exponential, correction polynomials, $q_j(1)$, partial sums
U_n, G_n, Π_n	unmatched count (Part II’s U), gap from maximum size, conditioned Poisson law
$N(y), F_J, x_J, \varepsilon_J$	threshold, smooth model, its inverse, bracket half-width
ℓ, w, x_0, D, C	$\log y, W(4\ell/(3e))$, initializer, $H'(x_0)$, $\log(3\sqrt{2}) - 3/8$
J, M	truncation orders (Part II’s J is an edge count)

1 Counting convention and attribution

The three parts of $K_{n,n,n}$ are distinguished and contain labeled vertices. A matching is a set of edges with disjoint endpoints. It is *maximal* if no edge can be added, and *maximum* if its cardinality is largest possible. These are different conditions. We count matchings without identifying graph

automorphisms. Set $A_0 = 1$ for the empty graph; asymptotic assertions concern positive integers $n \rightarrow \infty$.

The live OEIS entry A297487 [7], consulted on 3 October 2026, records the leading equivalent

$$A_n \sim 3\sqrt{2} \left(\frac{2n}{e}\right)^{3n/2} \exp\left(\sqrt{\frac{n}{2}} - \frac{3}{8}\right), \quad (1)$$

credited to Václav Kotěšovec on 6 February 2026. The entry also gives Andrew Howroyd’s positive-sum program dated 30 December 2017 and a hypergeometric expression containing the even perfect-matching correction. Neither the leading equivalent, the positive sum, nor that exact correction is claimed as new here. We supply a proof and refinements, using standard gamma asymptotics and Poisson moments.

[write] 6 October 2026. The write read the entry again (revision #18 of 6 February 2026); it is quoted in the front matter (“The OEIS entry, and the boundary sequence”). Kotěšovec’s equivalent there, $3 \cdot 2^{(3n+1)/2} e^{\sqrt{n/2} - 3n/2 - 3/8} n^{3n/2}$, is (1). The hypergeometric expression is the entry’s Mathematica line, which carries no name; Howroyd’s program is (5) at $v = 1$, term by term (proved in the front matter). The entry’s author is Eric W. Weisstein (30 December 2017); the terms from $a(6)$ on are Howroyd’s.

A bounded literature search found related Hermite pairing theory [1, 8], extremal cardinalities of maximal matchings in complete multipartite graphs [10], and enumeration over subcritical graph classes [4]. Paris’s full paper was read; the Azor–Gillis–Victor and Song–Miao–Wang–Zhao full texts were not obtained. Their abstracts and available primary metadata do not certify theorem-by-theorem nonoverlap. No matching marked all-fixed-order result was located in the sources examined, but this is not a proof of worldwide novelty. Section 11 records the limits of this comparison.

Throughout put

$$\lambda = \sqrt{n/2}, \quad B_n = \frac{(n!)^3}{\Gamma(n/2 + 1)^3}, \quad h = \lambda^{-1}. \quad (2)$$

For a polynomial series, $[h^j]$ denotes formal coefficient extraction. Every use of “all orders” below means: for each fixed nonnegative integer truncation order, with constants allowed to depend on that order. It does not assert convergence of an infinite asymptotic series.

2 Exact enumeration and the marked polynomial

Lemma 2.1. *A matching of a complete multipartite graph is maximal if and only if all its unmatched vertices lie in at most one part.*

Proof. Two unmatched vertices in different parts could be joined by an edge. Conversely, if all unmatched vertices lie in one part, there is no edge between them and no edge can be added to the matching. $\square$

Suppose that k vertices in a distinguished part are unmatched. If a, b, c are the numbers of edges between the three pairs of parts, then

$$a + b = n - k, \quad a + c = n, \quad b + c = n.$$

Thus $a = b = (n - k)/2$, $c = (n + k)/2$. Admissible values of k are exactly

$$0 \leq k \leq n, \quad k \equiv n \pmod{2}.$$

Partition the vertices of each part according to their destinations, and then bijectively pair the classes assigned to each edge type. The resulting count is

$$t(n, k) = \frac{(n!)^3}{k! ((n-k)/2)!^2 ((n+k)/2)!}. \quad (3)$$

For $k > 0$ the unmatched part is unique. At $k = 0$ the three choices represent the same perfect matching. Define

$$S_n(v) = \sum_{\substack{0 \leq k \leq n \\ k \equiv n \pmod{2}}} t(n, k) v^k. \quad (4)$$

If $A_n(v)$ counts maximal matchings with weight v^k for k unmatched vertices, then

$$A_n(v) = 3S_n(v) - 2\mathbf{1}_{2|n}B_n, \quad A_n = A_n(1). \quad (5)$$

Here $B_n = t(n, 0)$ when n is even; for odd n it is only a useful positive normalization. Formula (5) also gives $A_0(v) = 1$ directly, before introducing λ .

For part sizes a, b, c , let $M_{a,b,c}(v)$ be the analogous marked count. The multivariate exponential generating function is

$$\sum_{a,b,c \geq 0} M_{a,b,c}(v) \frac{x^a y^b z^c}{a!b!c!} = (e^{vx} + e^{vy} + e^{vz} - 2)e^{xy+xz+yz}. \quad (6)$$

The first factor chooses the unmatched set in one part and counts its empty choice once; the second is a labeled set of cross-part edges. In particular $A_n(v) = (n!)^3 [x^n y^n z^n]$ of the right side.

The first values for $n = 0, \dots, 8$ are

$$1, 3, 14, 342, 5256, 252360, 7950960, 582346800, 30400755840.$$

The package checks all 16 displayed OEIS values and independently computes $n = 0, \dots, 100$. The remote b-file was not retrieved, so a full b-file comparison is not claimed.

[write] **6 October 2026.** The write retrieved the b-file (Howroyd, $n = 1, \dots, 100$) and compared it with the 101 exact counts of the package's certificate (`data/176-balanced-data-certificates.json`): all 100 terms agree. The two sentences above are kept as the Part's statement of what it checked. The write also recomputed the nine values displayed above by coefficient extraction from (6).

3 A globally bounded Poisson reduction

Set $x = \lambda^2 = n/2$ and define, for integers $0 \leq k \leq n$,

$$R_n(k) = \frac{\Gamma(x+1)^3}{\Gamma(x-k/2+1)^2 \Gamma(x+k/2+1) \lambda^k}. \quad (7)$$

Then $t(n, k) = B_n \lambda^k R_n(k)/k!$ on the admissible parity class.

Lemma 3.1. *For $n \geq 1$ and every integer $0 \leq k \leq n$,*

$$0 < R_n(k) \leq e^2. \quad (8)$$

Consequently, for $P \sim \text{Poisson}(v\lambda)$ and real $v > 0$,

$$\frac{S_n(v)}{B_n e^{v\lambda}} = \mathbb{E} \left[R_n(P) \mathbf{1}_{P \leq n} \mathbf{1}_{P \equiv n \pmod{2}} \right]. \quad (9)$$

The product with the cutoff is understood as zero when $P > n$.

Proof. Write $\psi = (\log \Gamma)'$. Convexity of $\log \Gamma$ and the functional equation imply

$$\psi(t) \leq \log t \leq \psi(t+1) \leq \log(t+1) \quad (t > 0).$$

With $a = k/2$,

$$\begin{aligned} \log \frac{\Gamma(x+1)}{\Gamma(x-a+1)} &= \int_{x-a}^x \psi(t+1) dt \leq a \log(x+1), \\ \log \frac{\Gamma(x+1)}{\Gamma(x+a+1)} &= - \int_x^{x+a} \psi(t+1) dt \leq -a \log x. \end{aligned}$$

The first integral is still valid when $x-a=0$, since $\psi(t+1)$ is continuous there. Twice the first bound plus the second, minus $a \log x$, gives

$$\log R_n(k) \leq k \log(1 + 1/x) \leq k/x \leq 2.$$

Positivity follows from the positive gamma arguments. Equation (9) is the exact Poisson probability formula. $\square$

Fix a marking compact $K = [a, b] \subset (0, \infty)$ and the larger interval

$$K^* = [a/2, 2b]. \quad (10)$$

The positive margins are important. Uniform Poisson Chernoff bounds give

$$\mathbb{P}(P/\lambda \notin K^*) = O_K(e^{-c_K \lambda}) \quad (11)$$

for some $c_K > 0$, uniformly in $v \in K$. A fixed polynomial in P/λ can be inserted without changing the exponential nature of this bound. Also $2b\lambda < n = 2\lambda^2$ eventually. Lemma 3.1 therefore controls both tails and the hard cutoff in the following argument.

4 Uniform gamma expansions and the coefficient algorithm

On K^* write $z = k/\lambda$ and $D(h, z) = \log R_n(k)$. For each fixed J , Stirling's expansion gives

$$D(h, z) = -\frac{3z^2}{8} + \sum_{j=1}^J d_j(z) h^j + O_{K,J}(h^{J+1}), \quad (12)$$

uniformly with any prescribed finite number of z derivatives. The first four polynomials are

$$d_1(z) = \frac{z}{4} - \frac{z^3}{48}, \quad d_2(z) = \frac{3z^2}{16} - \frac{z^4}{64}, \quad (13)$$

$$d_3(z) = -\frac{z}{24} + \frac{z^3}{48} - \frac{z^5}{640}, \quad d_4(z) = -\frac{z^2}{16} + \frac{3z^4}{128} - \frac{z^6}{640}. \quad (14)$$

For clarity, a finite procedure defines all further d_j . In the formal Stirling expression

$$G(X) = \left(X + \frac{1}{2}\right) \log X - X + \frac{1}{2} \log(2\pi) + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)X^{2m-1}}, \quad (15)$$

substitute $X = h^{-2}$ and $X = h^{-2} \pm z/(2h)$, and take the coefficient of h^j in

$$3G(h^{-2}) - 2G(h^{-2} - z/(2h)) - G(h^{-2} + z/(2h)) + \frac{z}{h} \log h. \quad (16)$$

Here $G(X)$ is the expansion of $\log \Gamma(X + 1)$, and only finitely many Bernoulli terms contribute to a fixed coefficient. The leading term of (16) is $-3z^2/8$; logarithms cancel. Taylor expansion of $\log(1 \pm zh/2)$ performs the extraction using rational arithmetic.

The uniformity and differentiated remainder are standard consequences of the sectorial log-gamma expansion [2, §5.11]. Indeed, for z in a fixed complex neighborhood of K^* , both gamma arguments tend uniformly to positive infinity inside a sector bounded away from the negative real axis. Use a sufficiently long uniform Stirling expansion there, and then Cauchy's estimate on fixed-radius z discs. This justifies finitely many z derivatives; it does not prove a complex- v version of the marked theorem.

Define $p_j(z)$ by

$$\exp \left(\sum_{s \geq 1} d_s(z) h^s \right) = \sum_{j \geq 0} p_j(z) h^j, \quad p_0 = 1, \quad p_j = \frac{1}{j} \sum_{s=1}^j s d_s p_{j-s}. \quad (17)$$

Thus d_j has rational coefficients, degree at most $j + 2$, and parity j , while p_j has degree at most $3j$ and parity j .

Theorem 4.1 (Marked expansion). *There are rational polynomials $q_j(v)$, with $q_0 = 1$, degree at most $3j$, and parity j , such that for every fixed $J \geq 0$ and compact $K \subset (0, \infty)$,*

$$S_n(v) = \frac{B_n}{2} e^{v\lambda - 3v^2/8} \left(\sum_{j=0}^J q_j(v) \lambda^{-j} + O_{K,J}(\lambda^{-J-1}) \right) \quad (18)$$

uniformly for real $v \in K$. A finite coefficient algorithm is

$$q_j(v) = e^{3v^2/8} [h^j] \exp \left(\sum_{r \geq 2} \frac{v h^{r-1}}{r!} \partial_z^r \right) \left[e^{-3z^2/8} \exp \left(\sum_{s \geq 1} d_s(z) h^s \right) \right] \Big|_{z=v}. \quad (19)$$

The parameter v is held fixed during differentiation in z . The operator is a formal coefficient algorithm, not an asserted convergent infinite operator.

Proof. Using (11) and (8), restrict (9) to $P/\lambda \in K^*$. Uniform exponentiation of (12) gives

$$R_n(P) = \sum_{j=0}^J h^j f_j(P/\lambda) + O(h^{J+1}), \quad f_j(z) = e^{-3z^2/8} p_j(z).$$

Taylor-expand f_j at $z = v$ through degree $2(J - j) + 1$. Its remainder is bounded by a constant times $|P/\lambda - v|^{2(J-j)+2}$ on the central interval. Exact even centered Poisson moments show

$$\mathbb{E}|P/\lambda - v|^{2m} = O_{K,m}(\lambda^{-m}).$$

Hence the expectation of this remainder after multiplication by h^j is $O(h^{J+1})$. The Taylor polynomial can be averaged over the whole Poisson distribution because its omitted tail is exponentially small. The exact centered moment generating function is

$$\mathbb{E} e^{s(P/\lambda - v)} = \exp \left(\sum_{r \geq 2} \frac{v h^{r-1} s^r}{r!} \right),$$

which gives (19) for the unsigned expectation.

It remains to impose the parity. Exactly,

$$\mathbf{1}_{P \equiv n \pmod{2}} = \frac{1 + (-1)^{P-n}}{2}, \quad \mathbb{E}[(-1)^P(P)_r] = (-v\lambda)^r e^{-2v\lambda}.$$

Every finite Taylor polynomial is a polynomial in P , so its signed expectation is an exponential times a fixed polynomial in λ and λ^{-1} . The signed Taylor remainder has the same absolute bound as the unsigned one. Thus the parity-restricted expectation is half the unsigned expansion to every fixed algebraic order, proving (18).

For the degree assertion, an operator monomial of h order m uses ℓ derivative blocks, has total derivative order $r = m + \ell$, and carries a factor v^ℓ , where $\ell \leq m$. After extracting $e^{-3z^2/8}$, differentiation increases polynomial degree by at most r . Therefore it increases the eventual degree by at most $r + \ell \leq 3m$. Starting with p_{j-m} proves the bound $3j$. Derivative parity and the factor v^ℓ change parity by $r + \ell \equiv m \pmod{2}$, proving parity j . $\square$

Remark 4.2. The signed Taylor remainder above is bounded algebraically, not exponentially. The theorem therefore does not identify the exponentially small parity sector of the count. In particular, the exact subtraction in (5) cannot be promoted into an exponentially accurate count transseries by combining it with (18).

An alternative exact implementation of (19) uses the centered moments $\mu_r(t) = \mathbb{E}(P - t)^r$ for $P \sim \text{Poisson}(t)$:

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_{r+1}(t) = t\{\mu'_r(t) + r\mu_{r-1}(t)\}. \quad (20)$$

Only $r \leq 2J$ can affect order h^J . If $\mathcal{D} = \partial_z - 3z/4$, then $f_j^{(r)} = e^{-3z^2/8} \mathcal{D}^r p_j$, allowing every coefficient to be computed in rational polynomial arithmetic.

5 Four explicit corrections and the unmarked count

The first four polynomials obtained from Theorem 4.1 are

$$q_1(v) = \frac{v(25v^2 - 12)}{96}, \quad (21)$$

$$q_2(v) = \frac{v^2(625v^4 - 5208v^2 + 6768)}{18432}, \quad (22)$$

$$q_3(v) = \frac{v}{26542080} (78125v^8 - 1840500v^6 + 11037168v^4 - 16925760v^2 + 3525120), \quad (23)$$

$$q_4(v) = \frac{v^2}{10192158720} (1953125v^{10} - 90150000v^8 + 1295321760v^6 - 6818390784v^4 + 11911691520v^2 - 4349583360). \quad (24)$$

In particular, writing $a_j = q_j(1)$,

$$(a_0, a_1, a_2, a_3, a_4) = \left(1, \frac{13}{96}, \frac{2185}{18432}, -\frac{4125847}{26542080}, \frac{1950842261}{10192158720}\right). \quad (25)$$

For later use set

$$Q_J(v, \lambda) = \sum_{j=0}^J q_j(v) \lambda^{-j}, \quad Q_J(\lambda) = Q_J(1, \lambda). \quad (26)$$

On any fixed marking compact, $Q_J(v, \lambda)$ is positive for sufficiently large λ .

Corollary 5.1. *For each fixed $J \geq 0$,*

$$A_n = \frac{3}{2} B_n e^{\lambda-3/8} \left(Q_J(\lambda) + O_J(\lambda^{-J-1}) \right). \quad (27)$$

If L_n denotes the right side of (1), then

$$\begin{aligned} \frac{A_n}{L_n} = 1 + \frac{13}{96\lambda} - \frac{119}{18432\lambda^2} - \frac{4575127}{26542080\lambda^3} \\ + \frac{1879441301}{10192158720\lambda^4} + O(\lambda^{-5}). \end{aligned} \quad (28)$$

With $C = \log(3\sqrt{2}) - 3/8$, the logarithm is

$$\begin{aligned} \log A_n = \frac{3n}{2} \log \frac{2n}{e} + \lambda + C + \frac{13}{96\lambda} - \frac{1}{64\lambda^2} \\ - \frac{5243}{30720\lambda^3} + \frac{425}{2048\lambda^4} + O(\lambda^{-5}). \end{aligned} \quad (29)$$

Proof. At $v = 1$ the subtracted term in (5) is $O(e^{-\lambda})$ relative to the main term in (18). It is absorbed at every fixed order. Stirling's expansion further gives

$$\log B_n = \frac{3n}{2} \log \frac{2n}{e} + \log(2\sqrt{2}) - \frac{1}{8\lambda^2} + \frac{7}{960\lambda^6} + O(\lambda^{-10}). \quad (30)$$

Multiplying Q_4 by $\exp(-1/(8\lambda^2))$ yields (28). Taking the formal logarithm yields (29). These are exact rational coefficient operations, not numerical fits. $\square$

[write] **6 October 2026.** *Second route to the leading term.* Part II's Theorem 14.1 applies to the fixed balanced proportions $(1, 1, 1)$. With $p_1 = p_2 = p_3 = n$ its quantities are $a_i = n/2$, $B = (n!)^3 / \Gamma(n/2 + 1)^3 = B_n$, $\lambda_i = \sqrt{(n/2)^2 / (n/2)} = \lambda$ and $C_i = \frac{\lambda^2}{8} \cdot \frac{6}{n} = \frac{3}{8}$, and each branch sum S_i is $S_n(1)$. So (66) reads $S_n(1) = \frac{B_n}{2} e^{\lambda-3/8} (1 + O(n^{-1/2}))$, which is the case $J = 0$, $v = 1$ of (18), and its count, $\sum_i S_i - 2B$ for even n and $\sum_i S_i$ for odd n , is (5). Part II's proof uses the same digamma inequalities and Poisson tilting as Lemma 3.1; it is leading-order only, but uniform over every compact set of strictly triangular proportions.

6 Marked laws and directly inserted moments

Let U_n be the number of unmatched vertices in a uniformly chosen maximal matching. Its probability generating polynomial is exactly $A_n(v)/A_n$. From Theorem 4.1, for every fixed J and compact $K \subset (0, \infty)$,

$$\mathbb{E}_v U_n = e^{\lambda(v-1)-3(v^2-1)/8} \left(\frac{Q_J(v, \lambda)}{Q_J(1, \lambda)} + O_{K,J}(\lambda^{-J-1}) \right). \quad (31)$$

The error is inside the displayed exponential normalization. This is a real-parameter mod-Poisson type statement only; no uniform fixed complex neighborhood of $v = 1$ is asserted.

Differentiating an unspecified real remainder would not justify moment expansions. Instead insert factorial powers in the exact sum. If $r \geq 1$ is fixed and $P \sim \text{Poisson}(\lambda)$, the exact factorial-shift identity

$$\mathbb{E}[(P)_r f(P)] = \lambda^r \mathbb{E}f(P+r) \quad (32)$$

implies

$$\mathbb{E}(U_n)_r = \frac{3B_n e^{\lambda} \lambda^r}{A_n} \mathbb{E} \left[R_n(P+r) \mathbf{1}_{P+r \leq n} \mathbf{1}_{P+r \equiv n \pmod{2}} \right]. \quad (33)$$

The perfect term has no contribution because $(0)_{\underline{r}} = 0$.

A finite inserted coefficient algorithm follows from (19) by replacing every z inside its square bracket by $z + rh$, retaining $v = 1$ in the operator, and then evaluating at $z = 1$. In detail let

$$T_r(h) = e^{3/8} \exp \left(\sum_{m \geq 2} \frac{h^{m-1}}{m!} \partial_z^m \right) \left[e^{-3(z+rh)^2/8} \exp \left(\sum_{s \geq 1} d_s(z+rh) h^s \right) \right] \Big|_{z=1} \quad (34)$$

as a formal series. Then $T_0(h) = \sum a_j h^j$. Applying the same enlarged-central-interval proof to the shifted argument gives, for each fixed M ,

$$\mathbb{E}(U_n)_{\underline{r}} = h^{-r} \left(\frac{\sum_{j=0}^M [h^j] T_r(h) h^j}{Q_M(1/h)} + O_{M,r}(h^{M+1}) \right). \quad (35)$$

Changing parity in (33) changes only the signed negligible polynomial contribution. The hard cutoff and tails are controlled as before. Formula (35) proves direct moment expansions at every fixed order; one chooses M sufficiently large before cancellations.

Theorem 6.1. *The first two moments have the expansions*

$$\mathbb{E}U_n = \lambda - \frac{3}{4} + \frac{21}{32\lambda} - \frac{9}{32\lambda^2} + O(\lambda^{-3}), \quad (36)$$

$$\text{Var } U_n = \lambda - \frac{3}{2} + \frac{71}{32\lambda} - \frac{41}{16\lambda^2} + O(\lambda^{-3}). \quad (37)$$

Proof. Use (35) for $r = 1, 2$ through h^5 , and form $\text{Var } U_n = \mathbb{E}(U_n)_2 + \mathbb{E}U_n - (\mathbb{E}U_n)^2$. The error for the mean is then $O(h^5)$ and for the second factorial moment $O(h^4)$; the subsequent square and subtraction leave an error at most $O(h^4)$ before truncating the displayed series, so $O(h^3)$ in (36)–(37) is justified.

For a compact way to check the retained coefficients, the already justified finite formal calculation agrees with applying $v\partial_v$ and its square to the formal logarithm in (31). In that calculation

$$q_2(v) - \frac{1}{2}q_1(v)^2 = -\frac{v^2(16v^2 - 23)}{64}.$$

The leading correction $-3v^2/8$, together with this polynomial and $q_1(v)$, gives exactly the displayed values. This algebraic check is used only after the inserted-moment remainder proof. $\square$

7 Conditioned Poisson comparison and local limits

Let Π_n denote the distribution of a Poisson random variable of mean λ conditioned to have parity n . We use the convention $d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_k |\mu(k) - \nu(k)|$.

Theorem 7.1. *As $n \rightarrow \infty$,*

$$d_{\text{TV}}(\mathcal{L}(U_n), \Pi_n) = O(\lambda^{-1/2}) = O(n^{-1/4}), \quad \frac{U_n - \lambda}{\sqrt{\lambda}} \implies N(0, 1). \quad (38)$$

For every fixed $M > 0$, uniformly over integers $k \equiv n \pmod{2}$ with $|k - \lambda| \leq M\sqrt{\lambda}$,

$$\mathbb{P}(U_n = k) = \frac{2}{\sqrt{2\pi\lambda}} e^{-(k-\lambda)^2/(2\lambda)} (1 + o(1)). \quad (39)$$

The factor two is the lattice-span correction.

Proof. First ignore the multiplicity adjustment at $k = 0$, and let V_n have probabilities $t(n, k)/S_n(1)$ on the admissible support. It is the Π_n law tilted by $R_n(k)\mathbf{1}_{k \leq n}$. On, for example, $k/\lambda \in [1/2, 2]$, the smooth gamma expansion yields

$$|R_n(k) - e^{-3/8}| \leq C \left(\frac{|k - \lambda|}{\lambda} + \lambda^{-1} \right).$$

The probability of each parity under $\text{Poisson}(\lambda)$ is $(1 + (-1)^n e^{-2\lambda})/2$, bounded away from zero. Cauchy–Schwarz and Poisson variance therefore give a conditioned first absolute deviation $O(\sqrt{\lambda})$. Outside the central interval, Chernoff bounds and $R_n \leq e^2$ apply. Hence

$$\mathbb{E}_{\Pi_n} |R_n(P)\mathbf{1}_{P \leq n} - e^{-3/8}| = O(\lambda^{-1/2}).$$

The normalizing expectation tends to $e^{-3/8} > 0$. Normalizing a nonnegative tilt changes its total variation distance by at most a constant times this L^1 error, proving $d_{\text{TV}}(\mathcal{L}(V_n), \Pi_n) = O(\lambda^{-1/2})$. For odd n , $U_n = V_n$ in distribution. For even n , the weights at zero change from $3B_n$ to B_n when passing from $3S_n$ to A_n ; their total variation effect is $O(B_n/S_n) = O(e^{-\lambda})$ by (18). This proves the first assertion.

For completeness the conditioned-Poisson central limit theorem follows from its exact moment generating function. With $\epsilon_n = (-1)^n$ and every real t ,

$$\begin{aligned} & \mathbb{E} \left[e^{t(P-\lambda)/\sqrt{\lambda}} \mid P \equiv n \pmod{2} \right] \\ &= \exp \left(\lambda(e^{t/\sqrt{\lambda}} - 1) - t\sqrt{\lambda} \right) \frac{1 + \epsilon_n e^{-2\lambda e^{t/\sqrt{\lambda}}}}{1 + \epsilon_n e^{-2\lambda}}. \end{aligned} \quad (40)$$

For each fixed t this tends to $e^{t^2/2}$. Moment generating function convergence in a real neighborhood of zero proves the conditional central limit theorem. Total variation decreases under a measurable transformation, so the first part of (38) transfers it to U_n . This transfer is not used to infer moments; those were proved separately.

For the local assertion, $k > 0$ eventually in the stated window, so the exact probability is $3t(n, k)/A_n$. Uniform Stirling expansion of $k!$ gives the ordinary Poisson local Gaussian approximation there. Also $R_n(k) = e^{-3/8}(1 + o(1))$ uniformly, while $A_n \sim (3/2)B_n e^{\lambda-3/8}$. Substitution gives (39). $\square$

[write] 6 October 2026. Part II's Corollary 14.2 with $\delta_1 = \delta_2 = \delta_3 = 0$ makes the unmatched part uniform in the limit (each probability tends to $1/3$; the perfect matchings, of probability $O(e^{-\lambda})$ by Theorem 8.1, are conditioned away) and gives $(U - \sqrt{n/2})/(n/2)^{1/4} \Rightarrow \mathcal{N}(0, 1)$. Since $\sqrt{\lambda} = (n/2)^{1/4}$, this is the central limit theorem in (38), by a second route. Part II's Theorem 14.1 also gives the conditioned Poisson approximation of each branch, without a rate; the rate $O(n^{-1/4})$ and the local law (39) are this Part's alone.

The number of edges is $(3n - U_n)/2$, and the largest possible number is $\lfloor 3n/2 \rfloor$. The gap from maximum size is therefore exactly

$$G_n = \frac{U_n - (n \bmod 2)}{2}. \quad (41)$$

In particular its mean is $\lambda/2 + O(1)$ and its standard deviation is $\sqrt{\lambda}/2(1 + o(1))$. Thus the typical gap has order $\sqrt{n}$ and its fluctuations have order $n^{1/4}$.

8 Rare maximum matchings

For even n , maximum matchings have $U_n = 0$; for odd n , they have $U_n = 1$. These are the smallest allowed unmatched counts, and (3) proves that they occur.

Theorem 8.1. *For every fixed $J \geq 0$, the probability $p_n^{\max}$ that a uniform maximal matching is maximum satisfies*

$$p_n^{\max} = \frac{2}{3}e^{-\lambda+3/8} \left(Q_J(\lambda)^{-1} + O_J(\lambda^{-J-1}) \right), \quad n \text{ even}, \quad (42)$$

$$p_n^{\max} = 2\lambda e^{-\lambda+3/8} R_n(1) \left(Q_J(\lambda)^{-1} + O_J(\lambda^{-J-1}) \right), \quad n \text{ odd}. \quad (43)$$

The fixed-shift ratio has an expansion to every fixed order in λ^{-2} , beginning

$$R_n(1) = 1 - \frac{1}{8\lambda^2} + \frac{17}{128\lambda^4} - \frac{75}{1024\lambda^6} + \frac{1035}{32768\lambda^8} + O(\lambda^{-10}). \quad (44)$$

These are relative expansions of exponentially small probabilities.

Proof. The exact numerators are B_n and $3t(n, 1) = 3B_n\lambda R_n(1)$, respectively. Divide by (27), using eventual positivity of Q_J . The relative error in the denominator remains a relative error in the quotient regardless of the small numerator.

For the odd ratio, the gamma functional equation gives, with $x = \lambda^2$,

$$R_n(1) = \frac{[\Gamma(x+1)/\Gamma(x+1/2)]^3}{(x+1/2)\sqrt{x}}. \quad (45)$$

The standard fixed-shift log-gamma expansion, obtained by substituting into Stirling's formula, gives for any fixed M

$$\log R_n(1) = \sum_{m=1}^M \ell_m x^{-m} + O_M(x^{-M-1}), \quad (46)$$

where the Bernoulli-polynomial formula is

$$\ell_m = \frac{3(-1)^{m+1}\{B_{m+1}(1) - B_{m+1}(1/2)\}}{m(m+1)} + \frac{(-1)^m}{m2^m}. \quad (47)$$

The first four values are $-1/8, 1/8, -11/192, 1/64$. Exponentiation gives (44) and all further fixed orders. $\square$

For explicit use, the four relative corrections after removing the leading prefactors are

$$\begin{aligned} \frac{p_n^{\max}}{(2/3)e^{-\lambda+3/8}} &= 1 - \frac{13}{96\lambda} - \frac{1847}{18432\lambda^2} + \frac{4912087}{26542080\lambda^3} \\ &\quad - \frac{2299743979}{10192158720\lambda^4} + O(\lambda^{-5}), \quad n \text{ even}, \end{aligned} \quad (48)$$

$$\begin{aligned} \frac{p_n^{\max}}{2\lambda e^{-\lambda+3/8}} &= 1 - \frac{13}{96\lambda} - \frac{4151}{18432\lambda^2} + \frac{5361367}{26542080\lambda^3} \\ &\quad - \frac{818433259}{10192158720\lambda^4} + O(\lambda^{-5}), \quad n \text{ odd}. \end{aligned} \quad (49)$$

Theorem 8.1 does not require an exponentially accurate expansion for A_n . Its exact rare numerator avoids subtracting nearly equal approximate quantities. This is the reason these probability results coexist with the limitation in Remark 4.2.

9 Smooth inversion and integer thresholds

Define the integer threshold

$$N(y) = \min\{n \geq 1 : A_n \geq y\}, \quad y > 0. \quad (50)$$

It exists because (1) tends to infinity.

Lemma 9.1. *The sequence A_n is nondecreasing, and is eventually strictly increasing.*

Proof. Given a maximal matching on the old vertices, add one new vertex to each part. If the old matching has unmatched vertices, leave the new vertex of that part unmatched and join the other two new vertices. If it is perfect, use the first distinguished part for the new unmatched vertex. All unmatched vertices of the result lie in one part, so it is maximal. Restricting to the old vertices recovers the original matching, proving an injection.

By (29), the difference of consecutive logarithms is $\log A_{n+1} - \log A_n \sim (3/2) \log(2n) > 0$. Here even a leading log expansion with an $o(1)$ remainder suffices to bound the consecutive remainder difference. This proves eventual strict increase. $\square$

For each fixed $J \geq 0$ define the smooth positive model, for all sufficiently large real x ,

$$F_J(x) = \frac{3}{2} \frac{\Gamma(x+1)^3}{\Gamma(x/2+1)^3} e^{\sqrt{x/2}-3/8} Q_J(\sqrt{x/2}). \quad (51)$$

Its logarithmic derivative is

$$\begin{aligned} \frac{d}{dx} \log F_J(x) &= 3\psi(x+1) - \frac{3}{2}\psi(x/2+1) + \frac{1}{4\sqrt{x/2}} \\ &\quad + \frac{Q'_J(\sqrt{x/2})}{4\sqrt{x/2} Q_J(\sqrt{x/2})} = \frac{3}{2} \log(2x) + O_J(x^{-1/2}) > 0. \end{aligned} \quad (52)$$

Consequently F_J has an eventual real inverse, denoted $x_J(y)$.

Theorem 9.2 (Boundary-safe threshold enclosure). *For every fixed $J \geq 0$ there are constants $c_J > 0$ and y_J such that, for $y \geq y_J$, with $x = x_J(y)$ and*

$$\varepsilon_J(y) = c_J \frac{x^{-(J+1)/2}}{\log x}, \quad (53)$$

one has

$$\lceil x - \varepsilon_J(y) \rceil \leq N(y) \leq \lceil x + \varepsilon_J(y) \rceil. \quad (54)$$

Eventually this integer bracket has width zero or one. It is an exact asymptotic existence statement, with no numerical error constant or onset certified here.

Proof. Write $a = (J+1)/2$ and $f_J = \log F_J$. The relative expansion (27) gives

$$|\log A_n - f_J(n)| \leq C_J n^{-a}$$

for all sufficiently large integers n . In a fixed-width neighborhood of x , equation (52) bounds f'_J below by a positive multiple of $\log x$, while n^{-a} is comparable with x^{-a} . Choose c_J larger than the corresponding error-to-slope constant. At $n_+ = \lceil x + \varepsilon_J \rceil$ the mean value theorem gives $\log A_{n_+} \geq \log y$. At $n_- = \lceil x - \varepsilon_J \rceil - 1 < x - \varepsilon_J$ it gives $\log A_{n_-} < \log y$. Monotonicity proves (54). Since $2\varepsilon_J < 1$ eventually, its endpoints differ by at most one. $\square$

Away from integer boundaries, if the two ceilings agree they determine $N(y)$ exactly. A bare rule $N(y) = \lceil x_J(y) \rceil$ is not proved: an arbitrarily small real error can change the ceiling near an integer. The existential constants in Theorem 9.2 also prevent interpreting an uncalibrated numerical bracket as a certified one.

[write] **6 October 2026.** Theorem 9.2 is proved directly at the integers, from Lemma 9.1 and the relative error of the model F_J ; it is not an instance of the staircase theorem of the transseries volume [11], whose item (2) reaches the same conclusion when the two ceilings agree (front matter, “The inverses and the transseries volume”). Part II’s Theorem 21.2 is the same kind of statement for the boundary sequence.

9.1 An explicit Lambert W approximation

Let $\ell = \log y$ and use the positive real branch of Lambert W :

$$w = W\left(\frac{4\ell}{3e}\right), \quad x_0 = \frac{2\ell}{3w}, \quad D = \frac{3}{2}(w+1), \quad \lambda_0 = \sqrt{x_0/2}. \quad (55)$$

These quantities are used only for sufficiently large y , so $\ell, w > 0$.

Theorem 9.3. *For every fixed $J \geq 0$,*

$$x_J(y) = x_0 - \frac{\lambda_0 + C}{D} + \frac{1}{4D^2} - \frac{3}{8D^3} + O_J\left(\frac{1}{\sqrt{x_0}D}\right). \quad (56)$$

In particular,

$$x_J(y) = x_0 - \frac{\lambda_0 + C}{D} + O(D^{-2}), \quad (57)$$

which has $o(1)$ absolute real error.

Proof. Let $H(x) = (3x/2)\log(2x/e)$ and $g(x) = \sqrt{x/2} + C$. Equations (30) and (51) imply

$$\log F_J(x) = H(x) + g(x) + O_J(x^{-1/2}).$$

The identity $we^w = 4\ell/(3e)$ gives $H(x_0) = \ell$, and

$$H'(x_0) = D, \quad H''(x_0) = \frac{3}{2x_0}, \quad g'(x_0) = \frac{1}{4\lambda_0}.$$

Monotonicity and the mean value theorem first localize the solution to $x_0 + O(\sqrt{x_0}/D)$. The $O(x^{-1/2})$ term in the logarithm changes the solution by $O(1/(\sqrt{x_0}D))$. Taylor inversion of $H + g$ then yields

$$\delta = -\frac{g}{D} + \frac{g'g}{D^2} - \frac{H''g^2}{2D^3} + O\left(\frac{1}{\sqrt{x_0}D}\right),$$

with functions evaluated at x_0 . Terms involving C in the last two corrections fit the stated remainder. The remaining coefficients follow from

$$g'g = \frac{1}{4} + O(x_0^{-1/2}), \quad H''g^2 = \frac{3}{4} + O(x_0^{-1/2}).$$

For the Taylor remainder, $H'''(x) = O(x^{-2})$ and $g''(x) = O(x^{-3/2})$ throughout the localized interval. Substituting $\delta = O(\sqrt{x_0}/D)$ and dividing the resulting residual by the slope $\asymp D$ gives terms of size $O(1/(\sqrt{x_0}D^3))$ or smaller. The unretained substitutions in the second-order correction obey the same bound. This proves (56). Finally $D \rightarrow \infty$ and $D/\sqrt{x_0} \rightarrow 0$, yielding (57). $\square$

[write] **6 October 2026.** x_0 and $D = H'(x_0)$ are the factorial core of the transseries volume [11] with $\kappa_{\text{vol}} = \frac{3}{2}$, $d_{\text{vol}} = \frac{3}{2}(\log 2 - 1)$, and w is that proposition's v_{vol} (an instance). The three correction terms of (56) are the terms $m = 1, 2$ of the volume's perturbation-operator series with $\varrho_{\text{vol}} = g$ (an analogue: the error bound is this Part's own). Both are proved in the front matter.

The models (51), together with (19), specify inverse approximations at arbitrarily high fixed algebraic orders. Each x_J may be obtained by solving a one-dimensional eventually monotone equation. Formal Newton expansion about (55) gives further explicit terms when desired; the threshold theorem uses the model itself and does not require writing a long mixed expansion in $x_0^{-1/2}$ and D^{-1} .

10 Exact verification and reproducibility

The accompanying ZIP is an offline source and verification package, not a substitute for the uniform analytic proofs. Its standard-library Python code uses arbitrary precision integers and exact fractions. It reconstructs the gamma and Poisson coefficient calculations, checks the displayed count, logarithm, moment, and rare-probability coefficients, and compares exact finite counts. Two independent mathematical routes were used during development: shifted polygamma expansion with a cumulant operator, and Stirling substitution with centered Poisson moments. Their displayed coefficients agree exactly.

For the finite combinatorics, an independent recursion selects the first remaining labeled vertex. It either leaves that vertex unmatched, if this is compatible with the previously chosen unmatched part, or pairs it with each remaining vertex in a different part. At termination it records the total unmatched count. Memoization by remaining part counts and unmatched-part state avoids enumerating each labeled matching separately, while the number of available partners supplies the correct label multiplicity. Its full histograms agree with (3) through $n = 8$.

The certificates include exact totals for $0 \leq n \leq 100$; the 16 source-displayed values are distinguished from values computed by the formulas. Coefficient arrays, rational denominators, dimensions, parity, ranges, and schema are checked rather than trusted. Corruption tests run under both ordinary Python and `python -O`; mathematical validation does not depend on removable `assert` statements. Build and manifest checks reject changed, missing, added, or unsafe package entries.

The deterministic build uses a fixed source inventory, fixed timestamps, normalized PDF metadata, and `pdflatex -no-shell-escape`. The ZIP includes the article source, PDF, code, certificates, source audit, instructions, and an exact SHA-256 manifest. A rebuild into a new directory is required; existing output is not overwritten. Byte-for-byte reproducibility is tested in fresh extracted copies using the same installed TeX distribution. Cross-version TeX installations may produce different PDF bytes while preserving the mathematics and source integrity.

Finite tests do not prove uniform tail estimates, all-order remainder bounds, a numerical onset, or worldwide novelty. Those distinctions are deliberate. In particular, numerical stabilization of a rescaled omitted term is a diagnostic and is not used as an analytic proof.

[write] In this directory the package's programs are `code/176-balanced-*.py`; its certificate and receipts are `data/176-balanced-data-*.json` and `data/176-balanced-generated-*.json`; the frozen reference is `data/176-balanced-references-independent_results.json`; and its guides are `176-balanced-README_CODE.md` and `176-balanced-SOURCES.md`. `Report176.pdf`, the delivered README and `SHA256SUMS.json` are not shipped. The programs expect the delivered layout; the report's README gives the rerun route.

11 Source scope and further questions

11.1 What was compared

The source audit in the package gives links and the access level for each source. OEIS A297487 supplies the exact target, credited leading formula, and positive-sum program. Paris [8] gives an all-order expansion for four equal Hermite factors; this already covers a nearby perfect-matching problem and is a reason not to treat every apparently unexpanded OEIS entry as a fresh target. The present object is the marked count of maximal matchings on the fixed dense graph $K_{n,n,n}$.

Azor, Gillis, and Victor [1] develop Hermite-polynomial pairing applications; their full paper was unavailable in this search. Song, Miao, Wang, and Zhao [10] study maximum and minimum cardinalities and edge domination in complete multipartite graphs according to their primary abstract; their full paper was also unavailable. Drmota, Ramos, Requilé, and Rué [4] count graph-and-marking pairs over series-parallel and related graph classes, a different random model. These observations delimit the comparison; they do not exclude an unlocated theorem.

A bounded check of the accessible report catalogue and repository found no A297487 or maximal-tripartite asymptotic report. Search negatives and incomplete catalogues are supporting evidence only. The appropriate description of this article is a self-contained derivation and refinement of the credited OEIS equivalent, with no global first-priority assertion.

[write] 6 October 2026. This report is now that report. No other placed report treats A297487 or maximal matchings of complete multipartite graphs (repository search, 6 October 2026); Part II continues this Part to unequal parts.

11.2 Questions beyond the present theorems

1. **Exponentially accurate count parity.** Determine the signed gamma-weighted Poisson expectation beyond all algebraic orders. This would require a separate analysis of the signed remainder, not just the exact perfect-matching subtraction.
2. **Marking transition regimes.** The proof is uniform only when v stays in a compact subset of $(0, \infty)$. The scales $v \rightarrow 0$ and $v \rightarrow \infty$ may interpolate between fixed unmatched counts and macroscopic deficiency. The present theorem gives no uniform statement there.
3. **Complex marking and sharper local laws.** Establish a controlled complex- v expansion, if available, and derive Edgeworth corrections with explicit lattice terms. Real-positive uniformity alone is insufficient.
4. **Effective constants.** Quantify Stirling, tail, and Taylor remainders to produce certified finite- n errors and usable threshold onsets. Such work would turn the existential ceiling enclosure into a numerical certificate.
5. **Unequal parts and more parts.** For $K_{a,b,c}$ or a growing number of parts, identify the feasibility boundaries and competing unmatched-part contributions. Symmetry is central to the simple scalar gamma ratio used here.

[write] 6 October 2026. These five questions are items 1–5 of Section 26. Question 5 is *re-scoped*: Part II, written the same day, settles the instance $K_{2n+d,n,n}$ across the face $a = b + c$ of the triangle inequality (Theorems 15.1 and 22.1, with the suppression (75)) and identifies the competing branch contributions for compact strictly triangular proportions to leading order (Theorem 14.1). More parts, faces with unequal small parts, uniform transitions between the regimes and higher orders in the strict triangle remain open (front matter, “Part II and the question of Part I”).

Part II

Maximal matchings across the triangle boundary: boundary asymptotics, a uniform moving window, and an exterior Poisson law

Bundle Report 178, archive `Unequal_Tripartite_Matchings_Transitions_and_Inverses_Source.zip`, delivered as `Report178.tex` (19 pages). Delivered title block “Maximal Matchings Across the Triangle Boundary / Boundary asymptotics, a uniform moving window, and an exterior Poisson law / Research report 178 / 3 October 2026”. Printed as delivered, with `[write]` notes; Report 178’s Section k is Section $k + 11$ here, its statements move with their sections, and its equation (k) is equation $(k + 57)$. Its citation key `dlmf` is `dlmf16` here, and its reference `r176` is Part I.

Abstract of this Part (as delivered). We count uniform maximal matchings of labeled complete tripartite graphs, continuing the balanced $K_{n,n,n}$ analysis of Report 176 and OEIS A297487. The boundary sequence for $K_{2n,n,n}$ has no established OEIS identifier in this report. Its unmatched count is of order $n^{2/3}$, with Gaussian fluctuations of order $n^{1/3}$. We prove an all-fixed-orders count expansion, give four relative corrections, refine the moments, prove local and total-variation Gaussian laws, and obtain boundary-safe inverse enclosures. Exact maximum-matching numerators handle rare events separately from central approximations.

For $K_{2n+d,n,n}$, a single rational coefficient algorithm gives an all-fixed-orders expansion uniform when the exact parameter $d/n^{2/3}$ remains in a fixed real compact interval. The proof suppresses the other unmatched-part branches uniformly, including odd parity and the endpoint $d = 0$. For d/n in a positive compact interval, the deficit from a maximum matching instead has an all-orders Poisson expansion. A strictly triangular comparison and near-balanced softmax law identify the different geometric regimes.

These results use classical matching enumeration, gamma asymptotics, discrete Laplace expansion, and Poisson summation. No global novelty claim, effective error constants, unconditional rounding rule, or uniform matching as $|d|/n^{2/3} \rightarrow \infty$ is asserted. Reproducible exact checks accompany the proofs; ordinary high-precision diagnostics are distinguished from certified total numerical error.

Reading conventions of Part II `[write]`

$M(p_1, p_2, p_3), A(n, d), A_n$	count on K_{p_1, p_2, p_3} ; $M(2n + d, n, n)$; the boundary count $A(n, 0)$ (not A297487)
$a_i, T_i(k), S_i, B, \lambda_i, C_i$	edge-population offsets, branch terms, branch sums, gamma reference, branch means, curvature constants (at balance Part I’s B_n, λ and $3/8$)
$J, t(n, d, j), Z, U$	edges between the two parts of size n , the large-branch term and sum, the unmatched count ($d + 2J$ on the large branch)
N, q, a	$n^{1/3}, N^{-1}, 4^{-1/3}$
$\delta, u, s, \chi, v, b, \Phi, C$	window parameters (70)–(72) (v is a variance, not Part I’s marking variable)
$H(n, d), c_r(u), h_r, P_k, \mu_r$	normalizer, coefficients, phase polynomials, their exponential, Gaussian moments
c_k, ℓ_k, r_k	boundary coefficients in $\mathbb{Q}(a)$
$B_M, x_M, H_*(y), \Delta_M$	smooth boundary model, its inverse, threshold, bracket half-width
$c, \lambda, R_j, \kappa_r, T_0$	$d/n, c^{-2}$, exterior ratio, coefficients, the $j = 0$ term (Section 22)
w, M	marking variable of (58) (Part I’s v); truncation order

12 Scope and counting convention

Every vertex is labeled and the three parts are distinguished. A matching is *maximal* when no edge can be added, and *maximum* when its cardinality is largest possible. The random object below is chosen uniformly from the set of maximal matchings. No random greedy matching process is involved.

Write $M(p_1, p_2, p_3)$ for the count on K_{p_1, p_2, p_3} , and set

$$A(n, d) = M(2n + d, n, n), \quad A_n = A(n, 0).$$

Exact identities allow nonnegative integer part sizes, with the empty graph counted once. Asymptotic statements concern positive integer $n \rightarrow \infty$; the indicated constraints ensure nonnegative part sizes eventually. We use

$$N = n^{1/3}, \quad q = N^{-1}, \quad a = 4^{-1/3}.$$

Every “all-orders” statement is a Poincaré expansion: each fixed truncation order has its own error constant and onset. Convergence of the infinite expansion is not claimed.

The balanced sequence $M(n, n, n)$ is A297487 [7]; it is not the boundary sequence A_n . Report 176 [9] gives the general exact generating function used here and leaves unequal parts and feasibility boundaries as further questions. The reductions in Section 13 are direct consequences of that framework and classical multipartite matching enumeration. The shifted Hermite identity is a classical generating-function consequence, not a claim of a new Hermite formula. Source comparison and unresolved full-text gaps appear in Section 25.

[write] **6 October 2026.** Report 176 is Part I of this report, and [9] refers to it. Its general generating function is (6), and the further question meant here is Part I’s question 5 (Section 11.2); what this Part settles of it is stated in the front matter (“Part II and the question of Part I”). The symbol A_n of this Part is the boundary count, not Part I’s A_n .

13 Exact enumeration and classical identities

Lemma 13.1. *A matching of a complete multipartite graph is maximal if and only if its unmatched vertices lie in at most one part.*

Proof. Two unmatched vertices in different parts can be joined by an edge. Conversely, if all unmatched vertices lie in one part, no edge joins two of them, so no edge can be added. $\square$

Let $M_{p_1, p_2, p_3}(w)$ weight a matching by w to the number of unmatched vertices. The multivariate exponential generating function is

$$\sum_{p_1, p_2, p_3 \geq 0} M_{p_1, p_2, p_3}(w) \frac{x^{p_1} y^{p_2} z^{p_3}}{p_1! p_2! p_3!} = (e^{wx} + e^{wy} + e^{wz} - 2) e^{xy + xz + yz}. \quad (58)$$

Indeed the second factor is the labeled set of cross-part edges. The first chooses the unmatched set in a single part; subtraction of two counts its empty choice once. This is the general EGF already recorded in Report 176.

[write] Lemma 13.1 is Part I’s Lemma 2.1, with the same proof, and (58) is Part I’s (6) with w for Part I’s v . Both are printed again because this Part’s proofs cite them.

For distinct i, j, ℓ put

$$a_i = \frac{p_j + p_\ell - p_i}{2}, \quad F = p_1! p_2! p_3!.$$

If exactly k vertices are unmatched in part i , the edge populations opposite the three parts are $a_i + k/2$, $a_j - k/2$, $a_\ell - k/2$. Consequently the exact branch term is

$$T_i(k) = \frac{F}{k! \Gamma(a_i + k/2 + 1) \Gamma(a_j - k/2 + 1) \Gamma(a_\ell - k/2 + 1)}. \quad (59)$$

This expression is used only when $k \geq 0$ and all three edge populations are nonnegative integers. Equivalently the feasibility constraints include $k \equiv p_1 + p_2 + p_3 \pmod{2}$. To prove the count, partition the vertices of each part by their destinations and unmatched class, and choose a bijection for each of the three edge types. The factorials cancel to (59). Positive- k branches are disjoint. A feasible perfect term occurs in all three branch sums and must be subtracted twice.

13.1 The large-part branch

Let J denote the number of edges joining the two parts of size n . On the branch whose unmatched vertices are in the part of size $2n + d$, the three edge populations are $n - J, n - J, J$ and the unmatched count is $U = d + 2J$. Therefore

$$A_{\text{big}}(n, d) = \Gamma(2n + d + 1) Z(n, d), \quad (60)$$

$$Z(n, d) = \sum_{j=L}^n t(n, d, j), \quad L = \max(0, \lceil -d/2 \rceil), \quad (61)$$

$$t(n, d, j) = \frac{\Gamma(n + 1)^2}{\Gamma(n - j + 1)^2 \Gamma(d + 2j + 1) \Gamma(j + 1)}. \quad (62)$$

For $L \leq j < n$ its positive-term ratio is

$$r(n, d, j) = \frac{(n - j)^2}{(d + 2j + 2)(d + 2j + 1)(j + 1)}. \quad (63)$$

It is strictly decreasing. If $d > 0$, all vertices of the large part cannot be matched outside it, so there are no other branches. If $d = 0$, the other two formal branches contain only the common perfect term, which their duplicate correction cancels. Thus $A(n, d) = A_{\text{big}}(n, d)$ for $d \geq 0$. For $d < 0$ this equality generally fails; Section 16 proves an exponentially small relative difference in the moving window.

13.2 The exact boundary sequence

For $d = 0$,

$$A_n = (2n)! Z_n, \quad Z_n = \sum_{j=0}^n \frac{(n!)^2}{(2j)!(n - j)!^2 j!}, \quad U = 2J. \quad (64)$$

The edge count is $2n - J$. The $j = 0$ term is 1, so precisely $(2n)!$ matchings are maximum, and each of those is perfect. The prefix for $n = 0, \dots, 10$ is

$$\begin{aligned} &1, 3, 74, 4506, 489240, 82306920, \\ &19743705360, 6381638963280, 2667409223310720, \\ &1397546505279388800, 895714536024011692800. \end{aligned}$$

No OEIS identifier is assigned to this prefix.

[write] **6 October 2026.** The write searched the OEIS for 74,4506,489240,82306920 and for 4506,489240,82306920,19743705360: “No results”. It recomputed the eleven displayed values by coefficient extraction from (58) at $(2n, n, n)$, independently of (64); they agree.

Two useful exact identities are

$$Z_n = {}_2F_2\left(\begin{matrix} -n, -n \\ 1/2, 1 \end{matrix}; \frac{1}{4}\right), \quad A_n = \mathbb{E}\left[\text{He}_{2n}(X+1) \text{He}_n(X)^2\right], \quad (65)$$

where $X \sim \mathcal{N}(0, 1)$ and He denotes probabilists’ Hermite polynomials. The first follows from $(1/2)_j = (2j)!/(4^j j!)$. For the second, the product of the three Hermite generating functions has expectation

$$\mathbb{E} \exp\{(X+1)t - t^2/2 + Xs - s^2/2 + Xu - u^2/2\} = e^{t+ts+tu+su}.$$

Its $t^{2n}s^nu^n$ coefficient, multiplied by $(2n)!(n!)^2$, counts the large-part branch; at the boundary that is the entire count. This directly connects the problem with classical inhomogeneous-pairing theory [1, 8]. Neither identity justifies using a fixed-parameter large-argument hypergeometric formula for this moving terminating-parameter limit.

14 Strictly triangular proportions and competing parts

This comparison is proved independently of the boundary theorem. Suppose that the part ratios $(p_1/n, p_2/n, p_3/n)$ range in a fixed compact subset of the positive strict triangle region. Thus each a_i is bounded above and below by positive multiples of n . Define

$$B = \frac{F}{\prod_i \Gamma(a_i + 1)}, \quad \lambda_i = \sqrt{\frac{a_j a_\ell}{a_i}}, \quad C_i = \frac{\lambda_i^2}{8} \sum_{r=1}^3 a_r^{-1}.$$

When total parity is odd, B is a positive gamma reference, not a count of perfect matchings. Let S_i be the feasible branch sum including $k = 0$ when allowed.

Theorem 14.1. *Uniformly in such a compact strict triangle region,*

$$S_i = \frac{B}{2} \exp(\lambda_i - C_i) \{1 + O(n^{-1/2})\}. \quad (66)$$

The full count is $\sum_i S_i - 2B$ for even total size and $\sum_i S_i$ for odd total size. The subtraction, when present, is exponentially small relative to the sum. Within each nonempty branch the unmatched count is asymptotically, in total variation, a Poisson variable of mean λ_i conditioned on the required parity.

Proof. Write $T_i(k) = B\lambda_i^k R_i(k)/k!$ on its feasible range. The elementary digamma inequalities

$$\log t \leq \psi(t+1) \leq \log(t+1) \quad (t > 0) \quad (67)$$

follow from convexity of $\log \Gamma$ and its functional equation. Applying the upper bound to downward shifts and the lower bound to the upward shift gives

$$\log R_i(k) \leq \frac{k}{2} \{\log(1 + 1/a_j) + \log(1 + 1/a_\ell)\} \leq C.$$

The endpoint with a downward shift reaching zero is handled by continuity of $\psi(t+1)$ there. Thus the corresponding Poisson expectation has a uniform bounded tilt on its full feasible range.

For k in a fixed positive compact multiple of $\sqrt{n}$, Taylor expansion of $\log \Gamma$ gives

$$\log R_i(k) = -\frac{k^2}{8} \sum_r a_r^{-1} + O(n^{-1/2}). \quad (68)$$

For completeness, the leading linear gamma terms cancel $k \log \lambda_i$; the quadratic terms all have negative sign and sum to the displayed coefficient. Remaining linear terms are $O(k/n)$ and cubic terms are $O(k^3/n^2)$, both $O(n^{-1/2})$. The estimates and their first two derivatives hold uniformly on that central range.

Let $P \sim \text{Poisson}(\lambda_i)$, so $\lambda_i \asymp \sqrt{n}$. Outside a fixed central range its probability is $O(e^{-c\sqrt{n}})$ by the exponential-moment bound for a Poisson variable. The bounded tilt controls this tail, including the infeasible cutoff. On the central range, $R'_i(k) = O(n^{-1/2})$ and $R''_i(k) = O(n^{-1})$. Taylor's formula about λ_i has expected second-order remainder $O(n^{-1}\mathbb{E}(P - \lambda_i)^2) = O(n^{-1/2})$.

For either parity ϵ , $\mathbb{P}(P \equiv \epsilon \pmod{2}) = (1 + (-1)^\epsilon e^{-2\lambda_i})/2$, and

$$\mathbb{E}[(P - \lambda_i)\mathbf{1}_{P \equiv \epsilon \pmod{2}}] = -(-1)^\epsilon \lambda_i e^{-2\lambda_i}.$$

The linear Taylor term is therefore exponentially small even with parity. Using (68) at $k = \lambda_i$ proves (66). The perfect subtraction is smaller than $Be^{c\sqrt{n}}$ by an exponential factor. Finally $R_i(P)/R_i(\lambda_i) \rightarrow 1$ in probability on each parity class, and the global bound gives convergence in L^1 . Normalizing proves the total-variation assertion; excluding the exponentially rare $k = 0$ term has no effect. $\square$

Corollary 14.2 (Near-balanced softmax law). *Suppose $p_i = n + \delta_i \sqrt{n} + o(\sqrt{n})$ are integers, with fixed real δ_i , and put $D = \delta_1 + \delta_2 + \delta_3$. Condition away the perfect case, or give it its own label. The label I of the unmatched part has limiting probabilities*

$$\mathbb{P}(I = i) \longrightarrow \frac{e^{\sqrt{2}\delta_i}}{\sum_{r=1}^3 e^{\sqrt{2}\delta_r}}. \quad (69)$$

Jointly with this label,

$$\frac{U - \sqrt{n/2}}{(n/2)^{1/4}} \implies \mathcal{N}(0, 1),$$

and the limiting Gaussian and label are independent.

Proof. Expansion of a_i gives

$$\lambda_i = \sqrt{n/2} + \frac{4\delta_i - D}{2\sqrt{2}} + o(1), \quad C_i \longrightarrow 3/8.$$

The common factors in (66) cancel to give (69). Parity-conditioned Poisson variables with means tending to infinity have the usual centered Gaussian limit: their characteristic functions follow directly from $(e^{\lambda(z-1)} + (-1)^\epsilon e^{-\lambda(z+1)})/(1 + (-1)^\epsilon e^{-2\lambda})$. The branch means differ from $\sqrt{n/2}$ by only $O(1)$, negligible on the $n^{1/4}$ scale. The conditional Gaussian limit is the same for all three labels, proving joint independence. $\square$

The compact strict-triangle assumption is essential here. This proof supplies no estimate uniform as a triangle inequality degenerates.

[write] **6 October 2026.** At $p_1 = p_2 = p_3 = n$, Theorem 14.1 and Corollary 14.2 reduce to the leading term of Part I's Theorem 4.1 at $v = 1$ and to the central limit theorem of Part I's Theorem 7.1 (notes in Sections 5 and 7). Part I is all-order at balance; this Part is leading-order, but uniform over compact strict-triangle regions.

15 The compact moving-window theorem

Let $K \subset \mathbb{R}$ be any fixed compact interval. For each integer pair (n, d) put

$$\delta = \frac{d}{N^2} \in K.$$

All functions below use this *exact* parameter, not a rounded limit. There is a unique $u > 0$ satisfying

$$\delta = u - \frac{2}{u^2}. \quad (70)$$

The right side is strictly increasing from $-\infty$ to $+\infty$. Define

$$s = u^{-2}, \quad \chi = \frac{4}{u} + \frac{1}{s}, \quad v = \chi^{-1}, \quad b = -\frac{2s}{\chi}, \quad (71)$$

$$\Phi = u + s - \delta \log u, \quad C = -\frac{s^3}{3} + \frac{2s^2}{\chi}. \quad (72)$$

Equivalently $s(\delta + 2s)^2 = 1$ and $s > \max(0, -\delta/2)$, with $u = \delta + 2s$. These functions are real analytic in δ ; their positive quantities and reciprocals stay bounded on K . Set

$$H(n, d) = \frac{\Gamma(2n + d + 1)N^{-2d}}{N\sqrt{2\pi us\chi}} \exp\{N^2\Phi - s^2N + C\}. \quad (73)$$

Theorem 15.1. *For every fixed nonnegative integer M , there are rational functions $c_r(u)$, with $c_0 = 1$, such that uniformly for integer d with $d/N^2 \in K$,*

$$A(n, d) = H(n, d) \left\{ \sum_{r=0}^M c_r(u) N^{-r} + O_{K,M}(N^{-M-1}) \right\}. \quad (74)$$

They are analytic for $u > 0$, and Section 17 gives a finite coefficient algorithm. Moreover, for some positive γ_K ,

$$0 \leq \frac{A(n, d) - A_{\text{big}}(n, d)}{A_{\text{big}}(n, d)} \leq C_K e^{-\gamma_K N^2}. \quad (75)$$

The latter difference is exactly zero for $d \geq 0$.

We prove suppression next, then the complete local expansion and summation remainder. In particular this theorem covers $d = -1, -2, 0, 1$, both total parities, and all rounding offsets in the prescribed compact window.

16 Uniform suppression of the other branches

It remains to consider $d \leq 0$. Put

$$a_1 = -d/2, \quad a_2 = a_3 = n + d/2, \quad B = \frac{\Gamma(2n + d + 1)\Gamma(n + 1)^2}{\Gamma(a_1 + 1)\Gamma(a_2 + 1)\Gamma(a_3 + 1)}.$$

If d is odd, these are half-integer a_i and B is only a positive reference. All comparisons below use gamma functions, so this distinction causes no parity gap.

Suppose first that $a_1 > 0$. A small-part branch has $0 \leq k \leq 2a_1$ of the required parity. Its Poisson scale in (59) is $\lambda = \sqrt{a_1}$. The same integral inequalities (67) give

$$\frac{T_2(k)}{B} \leq \frac{\lambda^k}{k!} (1 + 1/a_1)^{k/2} (1 + 1/a_3)^{k/2} \leq \frac{\lambda^k}{k!} \exp\{k/(2a_1) + k/(2a_3)\}.$$

Here $k/(2a_1) \leq 1$, and $a_1/a_3 \leq 1$ uniformly for sufficiently large n . With $D = \max(0, -\inf K)$, summing even over all nonnegative k yields

$$S_2 + S_3 \leq 2e^2 B \exp(\sqrt{a_1}) \leq 2e^2 B \exp(\sqrt{D/2} N). \quad (76)$$

No lower bound proportional to N^2 on a_1 was used; the estimate includes $a_1 = 1/2$. At $a_1 = 0$, necessarily $d = 0$, the two small branches contain just their $k = 0$ terms. These contribute $2B$ and cancel exactly against the perfect duplicate correction.

To lower-bound the large branch, set $C_D = D/2 + 2$ and $c = 1/(8\sqrt{C_D})$. Take the largest nonnegative integer $k \leq cN^2$ with the required parity. For large N , uniformly,

$$k \geq cN^2/2, \quad a_2 - k/2 \geq n/2, \quad a_1 + k/2 + 1 \leq C_D N^2.$$

Its edge populations are nonnegative integers. The opposite directions of (67) give

$$\begin{aligned} \frac{T_1(k)}{B} &\geq \frac{(n/2)^k}{k!(a_1 + k/2 + 1)^{k/2}} \\ &\geq \left(\frac{1}{2c\sqrt{C_D}}\right)^k = 4^k \geq \exp(c \log 2 N^2). \end{aligned}$$

In the second line use $k! \leq k^k$ and $k \leq cN^2$. This construction works for odd total size as well. Combining it with (76) proves (75), after decreasing the positive exponential constant. Subtracting perfect duplicates can only reduce the nonnegative excess over the large branch. Since all statistics used here are polynomially bounded in n , every fixed polynomial moment transfers between the conditional large branch and the full model with superpolynomially small error.

17 Finite coefficient algorithm

For an admissible integer j , exactly

$$\log t = 2j \log n + 2 \sum_{h=0}^{j-1} \log(1 - h/n) - \log \Gamma(d + 2j + 1) - \log \Gamma(j + 1). \quad (77)$$

Set $j = sN^2 + Ny$ and $k = d + 2j = uN^2 + 2Ny$. Stirling expansion and the stationarity identity $\log s + 2 \log u = 0$ give the phase

$$\begin{aligned} \log t = & -2d \log N + N^2 \Phi - s^2 N - 2 \log N - \log(2\pi\sqrt{us}) \\ & - s^3/3 - 2sy - \chi y^2/2 + \sum_{r=1}^M h_r(y) N^{-r} + \text{remainder}. \end{aligned} \quad (78)$$

Completing the square yields

$$-s^3/3 - 2sy - \chi y^2/2 = C - (y - b)^2/(2v).$$

The first two phase polynomials are

$$h_1(y) = s - s^4/6 - (s^2 + 1/(2s) + 1/u)y - y^2 + (1/(6s^2) + 4/(3u^2))y^3, \quad (79)$$

$$h_2(y) = -s^5/10 - (2s^3/3)y + s^2/2 - sy^2 + y - (1/s + 1/u)/12 \\ + y^2/(4s^2) + y^2/u^2 - y^4/(12s^3) - 4y^4/(3u^3). \quad (80)$$

Here is an explicit finite prescription for every further coefficient. Let $S_m(x) = \sum_{h=0}^{x-1} h^m$ mean the Faulhaber polynomial, and put $x = s/q^2 + y/q$, $z = u/q^2 + 2y/q$. Expand formally

$$G(q, y) = 3y/q - x \log(1 + yq/s) - z \log(1 + 2yq/u) \\ - \frac{1}{2} \log(1 + yq/s) - \frac{1}{2} \log(1 + 2yq/u) \\ - 2 \sum_{m \geq 1} \frac{q^{3m}}{m} S_m(x) - \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} \{x^{1-2r} + z^{1-2r}\}. \quad (81)$$

Its initial terms are $-s^2/q - s^3/3 - 2sy - \chi y^2/2$; $h_r = [q^r]G$. The separately displayed $N^2\Phi$ and logarithmic prefactors in (78) are not part of G . To obtain $h_1, \dots, h_M$, retain only $m \leq M+2$ and Stirling terms with $4r-2 \leq M$, and Taylor-expand the logarithms only through degrees which can affect q^M . This is a finite formal operation, not a convergent representation of the untruncated phase.

Define polynomials and Gaussian moments by

$$P_0(y) = 1, \quad P_k(y) = \frac{1}{k} \sum_{r=1}^k r h_r(y) P_{k-r}(y), \quad (82)$$

$$\mu_0 = 1, \quad \mu_1 = b, \quad \mu_{r+1} = b\mu_r + rv\mu_{r-1}. \quad (83)$$

For $Y \sim \mathcal{N}(b, v)$, the count coefficients are

$$c_k(u) = \mathbb{E}P_k(Y). \quad (84)$$

Equations (81)–(84) involve only rational operations in u , since $s = u^{-2}$. Gaussian insertion $\mathbb{E}[Y^m P_k(Y)]$, followed by formal series division, supplies moment coefficients. This proves the rational form of the coefficients once the analytic remainder below establishes the expansion.

18 Localization and the all-orders remainder

18.1 A uniformly interior saddle

On the admissible real range the ratio derivative is

$$\frac{d}{dx} \log r(n, d, x) = -\frac{2}{n-x} - \frac{2}{d+2x+2} - \frac{2}{d+2x+1} - \frac{1}{x+1} < 0. \quad (85)$$

Choose $\eta > 0$ smaller than $\min_K s/2$ and $\min_K u/4$. On $|x - sN^2| \leq \eta N^2$, both x and $d+2x$ are positive constant multiples of N^2 , uniformly, and $n-x \asymp n$. Hence the derivative lies between $-C_K/N^2$ and $-c_K/N^2$. For bounded y ,

$$\log r(n, d, sN^2 + yN) = N^{-1}(-2s - \chi y) + O_K(N^{-2}).$$

It follows by monotonicity that an integer mode m satisfies

$$m = sN^2 + bN + O_K(1). \quad (86)$$

Summing logarithms of consecutive ratios on this bulk interval gives

$$t(n, d, j)/t(n, d, m) \leq C_K \exp\{-c_K(j - m)^2/N^2\}. \quad (87)$$

Conversely at least $c_K N$ terms within a fixed multiple of N of the mode are bounded below by a fixed multiple of the modal term. Thus $Z \geq c_K N t(n, d, m)$.

At either bulk endpoint the limiting ratio $1/[z(\delta + 2z)^2]$, with $z = x/N^2$, is separated from 1 uniformly. Monotonicity bounds both exterior tails by geometric series whose initial terms already lose $e^{-c_K N^2}$. This reaches all the way to the lower cutoff, even when the unmatched count there is 0 or 1. The cutoff is separated from the saddle by $uN^2/2 + O(1)$ when $\delta < 0$, and by sN^2 when $\delta \geq 0$.

For every fixed sufficiently small $\varepsilon > 0$, discarding $|y| > N^\varepsilon$ now loses less than any prescribed inverse power of N , also after insertion of a fixed polynomial in y . In the bulk this follows from $e^{-cN^{2\varepsilon}}$; outside it follows from the geometric bounds. The full support has $|y| = O_K(N^2)$, so fixed polynomial weights cost only powers of N .

18.2 A polynomial-weighted local error

On $|y| \leq N^\varepsilon$, both j and k are comparable to N^2 , and $j/n = O_K(N^{-1})$. Truncate $\log(1 - h/n)$ in (77) at degree $L = M + 2$. Its total remainder is bounded by

$$C \sum_{h=0}^{j-1} (h/n)^{L+1} \leq C j^{L+2}/n^{L+1} = O_{K,M}(N^{-M-1}).$$

The power sums are exact polynomials. Stirling's real-positive remainder, after sufficiently many Bernoulli terms, is $O_{K,M}(N^{-M-1})$ for both factorial arguments. Taylor expansion in y/N then gives

$$\log t = \log D_{n,d} - (y - b)^2/(2v) + \sum_{r=1}^M N^{-r} h_r(y) + E_{M,n,d}(y), \quad (88)$$

$$D_{n,d} = \frac{N^{-2d} e^{N^2 \Phi - s^2 N + C}}{2\pi \sqrt{us} N^2}, \quad (89)$$

$$|E_{M,n,d}(y)| \leq C_{K,M} N^{-M-1} (1 + |y|^{D_M}) \quad (90)$$

for some finite integer D_M . Every h_r has degree at most $r + 2$. For the fixed order choose $\varepsilon > 0$ smaller, if necessary, so all Taylor remainders and retained perturbations are bounded by a constant plus an arbitrarily small multiple of y^2 . For the retained terms this follows from

$$N^{-r} |y|^{r+2} = (|y|/N)^r y^2;$$

for (90) it follows by choosing ε relative to D_M . Any such positive ε still leaves superpolynomially small tails.

Taylor's integral remainder for exponentiation yields

$$t(n, d, j) = D_{n,d} e^{-(y-b)^2/(2v)} \sum_{r=0}^M N^{-r} P_r(y) + R_{M,n,d}(y), \quad (91)$$

$$|R_{M,n,d}(y)| \leq C_{K,M} D_{n,d} N^{-M-1} (1 + |y|^{E_M}) e^{-c_K y^2}. \quad (92)$$

One may obtain (92) by differentiating the finite perturbation polynomial in q and applying Taylor's integral formula; its derivatives are polynomials in y , while the exponential of the perturbation is absorbed in a weaker Gaussian. This explains why the error is polynomial-weighted Gaussian, rather than merely the supremum of a polynomial on a growing window. The latter weaker approach could introduce unwanted logarithmic factors. Multiplication by any prescribed polynomial in y preserves the needed form of (92).

18.3 Summation on the actual lattice

The y values lie on the mesh $j/N - sN$, of spacing $1/N$, with arbitrary fractional offset. For every fixed polynomial P , Poisson summation for a polynomial times a Gaussian gives

$$\begin{aligned} \frac{1}{N} \sum_{j \in \mathbb{Z}} P(j/N - sN) e^{-(j/N - sN - b)^2 / (2v)} \\ = \int_{\mathbb{R}} P(y) e^{-(y-b)^2 / (2v)} dy + O_{K,P}(N^A e^{-c_K N^2}). \end{aligned} \quad (93)$$

To see uniformity, its Fourier transform is a polynomial times a Gaussian; the nonzero dual lattice points have size proportional to N , and the offset contributes only a unit-modulus phase. The compact bounds on b, v make all constants uniform. No assertion that odd moments vanish on an uncentered discrete lattice is needed.

Extend the central Gaussian-polynomial sum to the full lattice, losing less than any inverse power. Summing (92) costs $O(N)$ because shifted-mesh sums of a fixed polynomial times a Gaussian are uniformly $O(N)$. Finally

$$\Gamma(2n + d + 1) D_{n,d} N \sqrt{2\pi v} = H(n, d).$$

Equations (84), (91), and (93), followed by the suppression result, prove Theorem 15.1 with its full stated remainder. The argument also proves all fixed polynomial-insertion expansions used below.

19 Window corrections and probability laws

Put $x = u^3$. The first relative coefficient is

$$c_1(u) = \frac{60x^4 + 315x^3 + 372x^2 + 80x - 64}{6u^8(x+4)^3}. \quad (94)$$

For the second logarithmic coefficient, write

$$\begin{aligned} P(x) = 5x^8 - 340x^7 - 900x^6 + 6056x^5 + 17240x^4 \\ - 4800x^3 - 30720x^2 - 7680x + 6144. \end{aligned}$$

Then

$$\ell_2(u) = c_2(u) - \frac{c_1(u)^2}{2} = -\frac{P(x)}{60u^{10}(x+4)^5}, \quad (95)$$

so

$$\log \frac{A(n, d)}{H(n, d)} = \frac{c_1(u)}{N} + \frac{\ell_2(u)}{N^2} + O_K(N^{-3}).$$

These expressions follow by substituting (79)–(80) into $c_1 = \mathbb{E}h_1(Y)$ and $c_2 = \mathbb{E}[h_2(Y) + h_1(Y)^2/2]$, and reducing with $s = u^{-2}$. The formulas provide checkable rational expressions rather than numerical fits.

Theorem 19.1. *For a uniform maximal matching of $K_{2n+d,n,n}$ in the compact window,*

$$\mathbb{E}J = sN^2 + bN + m_0(u) + O_K(N^{-1}), \quad (96)$$

$$\text{Var } J = vN^2 + v_1(u)N + O_K(1), \quad (97)$$

$$m_0(u) = -\frac{3u^6 + 7u^3 - 8}{(u^3 + 4)^3}, \quad v_1(u) = -\frac{4(u^6 + 2u^3 + 4)}{u(u^3 + 4)^3}. \quad (98)$$

For the actual unmatched count,

$$\mathbb{E}U = uN^2 + 2bN + 2m_0(u) + O_K(N^{-1}), \quad (99)$$

$$\text{Var } U = 4vN^2 + 4v_1(u)N + O_K(1). \quad (100)$$

Let g_δ be the $\mathcal{N}(b, v)$ density, and define the normalized lattice Gaussian on all integers by

$$Q_{n,d}(j) = \frac{g_\delta((j - sN^2)/N)}{\sum_{r \in \mathbb{Z}} g_\delta((r - sN^2)/N)}. \quad (101)$$

Then, uniformly,

$$\sup_{j \in \mathbb{Z}} \left| N\mathbb{P}(J = j) - g_\delta((j - sN^2)/N) \right| = O_K(N^{-1}), \quad (102)$$

$$d_{\text{TV}}(\mathcal{L}(J), Q_{n,d}) = O_K(N^{-1}). \quad (103)$$

The Kolmogorov distance from the law of $(J - sN^2 - bN)/(\sqrt{v}N)$ to $\mathcal{N}(0, 1)$ is $O_K(N^{-1})$. The same conclusion holds for $(U - uN^2 - 2bN)/(2\sqrt{v}N)$.

Proof. Polynomial insertion in (91), followed by normalization, gives all fixed orders of the moments of $y = (J - sN^2)/N$. At first order, Gaussian integration by parts gives

$$m_0 = v\mathbb{E}h'_1(Y), \quad v_1 = v^2\mathbb{E}h''_1(Y),$$

which evaluate to (98). Run the insertion expansion one order further before multiplying by N or N^2 , so the remainders in (96)–(97) follow without differentiation of an uncontrolled O term.

On the large branch $U = d + 2J$. When $d < 0$ this is *not* an identity on the full set of maximal matchings. Equation (75) and polynomial boundedness of J, U transfer the moments to the full model, proving (99)–(100).

The normalized first-order pointwise expansion, its Gaussian-weighted error, and the ratio tail estimates imply (102). Summing its weighted form gives total variation $O_K(N^{-1})$ against (101); the denominator is N up to an exponentially small relative error by (93). The full and conditional large-branch laws differ in total variation by at most the exceptional branch probability, so this is valid unconditionally. For U , couple it to $d + 2J$ on the same matching; equality fails only on that exceptional event.

Finally Riemann-sum approximation of a Gaussian distribution function on mesh $1/N$ has uniform error $O_K(N^{-1})$: bound each cell's error by the integral of the density derivative times its width, and use the finite uniformly bounded L^1 norm of that derivative. Combining this with the total-variation bound proves the Kolmogorov assertions. $\square$

If $\delta_n \rightarrow \tau$, the centering using s_n, b_n gives convergence of $(J - s_nN^2 - b_nN)/N$ to $\mathcal{N}(0, 1/\chi(\tau))$. Centering by the actual mean and standardizing by the actual variance gives the standard normal limit as well. These are statements with the exact δ_n retained in the centering; replacing it by its limit requires separate error control.

19.1 Why rounding cannot be dropped

The phase satisfies $\Phi'(\delta) = -\log u$. If $d = \tau N^2 + r_n$ with bounded r_n , then

$$N^2\{\Phi(d/N^2) - \Phi(\tau)\} = -r_n \log u(\tau) + O(N^{-2}).$$

Thus replacement by τ can introduce a nonvanishing factor $u(\tau)^{-r_n}$ in the phase contribution. The factors N^{-2d} and $\Gamma(2n + d + 1)$ also require the actual integer d . Formula (74) includes arbitrary rounding and parity automatically by using $\delta = d/N^2$ exactly.

20 The exact boundary and four corrections

At $d = 0$, the parameters simplify to

$$s = a, \quad u = 2a, \quad \chi = 3/a, \quad b = -2a^2/3, \quad v = a/3, \quad \Phi = 3a, \quad C = 1/12.$$

In particular the constant $1/12$ is forced by completing the square: the original constant is $-a^3/3$, and the shift contributes $2a^3/3$, leaving $a^3/3 = 1/12$.

Corollary 20.1. *For each fixed $M \geq 0$,*

$$A_n = \frac{(2n)!e^{3aN^2 - a^2N + 1/12}}{\sqrt{12\pi a} N} \left\{ \sum_{k=0}^M c_k N^{-k} + O_M(N^{-M-1}) \right\}. \quad (104)$$

Here $c_0 = 1$, and the first four relative coefficients are

$$\begin{aligned} c_1 &= \frac{211a}{216}, & c_2 &= \frac{172781a^2}{466560}, \\ c_3 &= -\frac{189855337}{1209323520}, & c_4 &= -\frac{6247138084769a}{36569943244800}. \end{aligned} \quad (105)$$

The first four coefficients of the logarithm of the relative series are

$$\ell_1 = \frac{211a}{216}, \quad \ell_2 = -\frac{173a^2}{1620}, \quad \ell_3 = -\frac{55}{324}, \quad \ell_4 = -\frac{389a}{122472}. \quad (106)$$

Proof. Specialize Theorem 15.1 and the coefficient algorithm at $u = 2a$, using $4a^3 = 1$. This yields (105) in the exact field $\mathbb{Q}(a)$. If $C(q) = 1 + \sum c_k q^k$ and $\log C(q) = \sum \ell_k q^k$, formal differentiation gives the finite recurrence

$$\ell_k = c_k - \frac{1}{k} \sum_{r=1}^{k-1} r \ell_r c_{k-r},$$

which gives (106). The all-fixed-order proof was established above, including the endpoint $d = 0$, so finite exact coefficient evaluation proves the displayed terms with the stated remainder. $\square$

Combining (106) with Stirling expansion of $(2n)!$ gives

$$\begin{aligned} \log A_n &= 2n \log(2n/e) + 3an^{2/3} - a^2n^{1/3} + \frac{1}{6} \log n + \frac{1}{12} - \frac{1}{2} \log(3a) \\ &\quad + \frac{211a}{216} n^{-1/3} - \frac{173a^2}{1620} n^{-2/3} - \frac{83}{648} n^{-1} - \frac{389a}{122472} n^{-4/3} + O(n^{-5/3}). \end{aligned} \quad (107)$$

The n^{-1} coefficient includes the factorial contribution $1/(24n)$; it is $-83/648$, not the relative-series logarithmic coefficient $-55/324$.

20.1 Boundary moments and a marked check

The insertion calculation through order four gives

$$\mathbb{E}J = aN^2 - \frac{2}{3}a^2N - \frac{1}{12} + \frac{49a}{162}N^{-1} - \frac{173a^2}{972}N^{-2} + \frac{1}{162}N^{-3} + O(N^{-4}), \quad (108)$$

$$\text{Var } J = \frac{a}{3}N^2 - \frac{4a^2}{9}N + \frac{1}{12} + \frac{17a}{243}N^{-1} - \frac{217a^2}{2916}N^{-2} + O(N^{-3}). \quad (109)$$

The mean uses normalized y moments through q^4 with error $O(q^5)$, then multiplies by N ; the variance uses the same orders and multiplies by N^2 . Thus these specific remainder orders require no differentiation of a bare asymptotic error. The exact identity $U = 2J$ gives its mean and variance by factors 2 and 4. In particular

$$\mathbb{E}U \sim 2^{1/3}n^{2/3}, \quad \sqrt{\text{Var } U} \sim 2\sqrt{a/3}n^{1/3}.$$

The local, total-variation, and Kolmogorov bounds of Theorem 19.1 specialize with $b = -2a^2/3$ and $v = a/3$. This gives both $(J - aN^2 - bN)/(\sqrt{v}N) \Rightarrow \mathcal{N}(0, 1)$ and the actual-mean, actual-variance standardized limit.

A useful independent algebraic check is obtained by marking j . For z in a fixed compact subset of $(0, \infty)$, define

$$Z_n(z) = \sum_{j=0}^n t(n, 0, j)z^j, \quad \alpha = (z/4)^{1/3}.$$

The same proof, uniform in this marking interval, replaces the boundary normalizer by

$$\frac{\exp(3\alpha N^2 - \alpha^2 N + \alpha^3/3)}{\sqrt{12\pi\alpha} N}.$$

Before imposing $\alpha^3 = 1/4$, the first coefficients are

$$\begin{aligned} c_1(\alpha) &= \alpha - 5\alpha^4/54, \\ c_2(\alpha) &= \frac{125\alpha^9 - 2196\alpha^6 + 8100\alpha^3 + 810}{29160\alpha}, \\ \ell_2(\alpha) &= \frac{28\alpha^6 - 360\alpha^3 + 45}{1620\alpha}. \end{aligned}$$

Applying the formal marking operator $z\partial_z = (\alpha/3)\partial_\alpha$ to the logarithmic model reproduces the displayed mean and variance coefficients at their applicable orders. The justification is the already proved polynomial-insertion expansion before taking ratios, not an assumption that arbitrary differentiation preserves an O term. Higher marking coefficients are generated by the same finite phase construction.

20.2 Maximum matchings are an exact rare event

By (64), the numerator for maximum matchings is exactly $(2n)!$. Hence

$$\mathbb{P}(\text{maximum}) = Z_n^{-1} = \sqrt{12\pi a} N e^{-3aN^2 + a^2N^{-1/12}} \left\{ \sum_{k=0}^M r_k N^{-k} + O_M(N^{-M-1}) \right\}, \quad (110)$$

where $r_0 = 1$ and

$$r_k = -\sum_{j=1}^k c_j r_{k-j}; \quad r_1 = -c_1, \quad r_2 = c_1^2 - c_2.$$

Event $J = 0$ lies far outside the central Gaussian scale. Formula (110) comes from its exact numerator and the positive count expansion, not from extrapolating the local limit theorem to a far tail.

21 Boundary-safe inverse enclosures

Lemma 21.1. $A_{n+1} \geq 2A_n$ for every $n \geq 0$.

Proof. Adjoin two labeled vertices to the large part and one to each small part. Match the new large vertices to the new small vertices in either of the two bijections. Either extension remains maximal by Lemma 13.1. The two images are disjoint, and restriction to old vertices recovers the old matching. $\square$

[write] This lemma concerns the boundary sequence $M(2n, n, n)$. Part I's Lemma 9.1 is the balanced counterpart (nondecreasing, by a different injection).

For fixed M define, for sufficiently large real x ,

$$B_M(x) = \frac{\Gamma(2x+1)e^{3ax^{2/3}-a^2x^{1/3}+1/12}}{\sqrt{12\pi a}x^{1/3}} \sum_{k=0}^M c_k x^{-k/3}. \quad (111)$$

The finite sum is eventually positive. Its logarithmic derivative is

$$(\log B_M)'(x) = 2\log(2x) + O(x^{-1/3}), \quad (112)$$

so it is eventually strictly increasing and unbounded. For sufficiently large y , let $x_M(y)$ be its unique large solution of $B_M(x) = y$ and let

$$H_*(y) = \min\{n \geq 0 : A_n \geq y\}.$$

Theorem 21.2. *There exist constants $C_M, y_M > 0$ such that for $y \geq y_M$, with $x_M = x_M(y)$ and*

$$\Delta_M = C_M \frac{x_M^{-(M+1)/3}}{\log x_M},$$

one has the boundary-safe enclosure

$$\lceil x_M - \Delta_M \rceil \leq H_*(y) \leq \lceil x_M + \Delta_M \rceil. \quad (113)$$

It eventually contains at most two consecutive integers.

Proof. Corollary 20.1 implies

$$|\log A_n - \log B_M(n)| \leq Cn^{-(M+1)/3}$$

for sufficiently large n . On a neighborhood of x_M , (112) is bounded below by a positive multiple of $\log x_M$. The mean-value theorem, with C_M enlarged, implies $A_n < y$ for every nearby integer $n < x_M - \Delta_M$, and $A_n > y$ for every nearby integer $n \geq x_M + \Delta_M$. Monotonicity extends these comparisons outside the neighborhood, and the finitely many small n are below y eventually. Taking ceilings proves (113). Since $2\Delta_M \rightarrow 0$, its endpoints are eventually equal or consecutive. $\square$

All constants and the onset here are existential. This is not a numerically certified inverse algorithm for arbitrary y , and no unconditional rule $H_*(y) = \lceil x_M(y) \rceil$ follows when x_M is near an integer.

The leading Lambert approximation is

$$x_0 = \frac{\log y}{2W((\log y)/e)},$$

which solves $2x_0 \log(2x_0/e) = \log y$. Its first displacement is

$$x_M - x_0 \sim -\frac{3ax_0^{2/3}}{2\log(2x_0)}.$$

Indeed the missing leading term in the logarithmic count is $3ax_0^{2/3}$, and the leading derivative is $2\log(2x_0)$; the relative displacement tends to zero, making the Taylor remainder smaller. This is only a leading displacement, not a complete ordered reversion series. Implicit inversion of (111) avoids mixing logarithmic and fractional-power scales.

[write] 6 October 2026. x_0 is the factorial core of the transseries volume [11] with $\kappa_{\text{vol}} = 2$, $d_{\text{vol}} = 2(\log 2 - 1)$ (an instance), and $2\log(2x_0)$ is its slope $\kappa_{\text{vol}}(v_{\text{vol}} + 1)$. The displayed displacement is the first term of the volume's perturbation-operator series with $\varrho_{\text{vol}}(x) = 3ax^{2/3}$ (an analogue). Theorem 21.2 is proved directly at the integers and is not an instance of the volume's staircase theorem. Details in the front matter.

22 The fixed exterior ratio and its Poisson expansion

Now suppose the exact ratio $c_n = d/n$ ranges in a fixed compact subset $K_+ \subset (0, \infty)$. Write $c = c_n$ and $\lambda = c^{-2}$. Only the large branch is possible. The $j = 0$ term including its factorial carrier is

$$T_0 = \frac{\Gamma(2n + d + 1)}{\Gamma(d + 1)}.$$

Relative to it, the terms are

$$R_j = \frac{(n)_j^2 \Gamma(d + 1)}{j! \Gamma(d + 2j + 1)}, \quad 0 \leq j \leq n, \quad A(n, d) = T_0 \sum_{j=0}^n R_j. \quad (114)$$

Here $(n)_j = n(n-1) \cdots (n-j+1)$, with empty product 1. In particular,

$$0 \leq R_j \leq \frac{(n/d)^{2j}}{j!} = \frac{\lambda^j}{j!}. \quad (115)$$

Theorem 22.1. *For every fixed $M \geq 0$, rational functions $\kappa_r(c)$ with $\kappa_0 = 1$ satisfy, uniformly on K_+ ,*

$$A(n, d) = T_0 e^\lambda \left\{ \sum_{r=0}^M \kappa_r(c) n^{-r} + O_{K_+, M}(n^{-M-1}) \right\}. \quad (116)$$

In particular,

$$\kappa_1(c) = -(c^{-4} + 2c^{-5} + 3c^{-3}), \quad d_{\text{TV}}(\mathcal{L}(J), \text{Poisson}(c^{-2})) = O_{K_+}(n^{-1}). \quad (117)$$

Here $U = d + 2J$ exactly. A maximum matching has $2n$ edges; J is exactly its edge-count deficit. If $c_n \rightarrow c_ > 0$, then $J \Rightarrow \text{Poisson}(c_*^{-2})$ in total variation and $U - d$ converges to twice that Poisson variable.*

Proof. For fixed j , finite products in (114) give $R_j/(\lambda^j/j!) \rightarrow 1$. Bound (115) already proves the leading count and total-variation convergence by dominated summation. To prove the full expansion

and rate, write

$$\begin{aligned} \log \frac{R_j}{\lambda^j/j!} &= 2 \sum_{h=0}^{j-1} \log(1 - h/n) - \sum_{h=1}^{2j} \log(1 + h/(cn)) \\ &= \sum_{m=1}^M \beta_m(j; c) n^{-m} + O_{K_+, M}(n^{-M-1}(1 + j^{M+2})), \end{aligned} \quad (118)$$

$$\beta_m(j; c) = -\frac{2}{m} S_m(j) + \frac{(-1)^m}{m c^m} \sum_{h=1}^{2j} h^m. \quad (119)$$

These are exact coefficient polynomials in j ; the remainder is uniform on $j \leq n^\varepsilon$ for a sufficiently small fixed positive ε . Define $E_0 = 1$ and

$$E_r(j; c) = \frac{1}{r} \sum_{m=1}^r m \beta_m(j; c) E_{r-m}(j; c).$$

Then a finite exponential Taylor expansion gives a polynomial-weighted remainder of order n^{-M-1} on this central range. Choose ε small enough to keep the retained perturbation bounded there. Every resulting polynomial may be summed against $\lambda^j/j!$. The complementary tail $j > n^\varepsilon$ is smaller than every fixed inverse power, also after any fixed polynomial insertion, because λ remains in a fixed positive compact interval and (115) holds globally. Thus

$$\kappa_r(c) = \mathbb{E}[E_r(P; c)], \quad P \sim \text{Poisson}(c^{-2}), \quad (120)$$

proves (116). Poisson moments are finite polynomials in λ (for example via factorial moments $\mathbb{E}(P)_k = \lambda^k$), so the algorithm is rational and finite.

The first phase coefficient is $\beta_1 = -j(j-1) - j(2j+1)/c$. Taking Poisson expectation gives (117). The same weighted expansion of the masses, followed by normalizing and summing absolute errors, gives total variation $O(n^{-1})$. Finally the graph has only $2n$ vertices outside the large part, so no matching has more than $2n$ edges, and the $j = 0$ term attains that number. This proves the deficit interpretation. $\square$

The maximum-matching probability here also has an exact numerator, T_0 , so its expansion is the reciprocal of the factor multiplying T_0 in (116). In particular it equals

$$e^{-\lambda} \{1 + (c^{-4} + 2c^{-5} + 3c^{-3})/n + O_{K_+}(n^{-2})\}.$$

It is no longer exponentially small in n when c stays in a positive compact interval.

23 Regimes and what is not matched

The different scales can be summarized without extending any uniformity claim.

Assumption	Unmatched count	Fluctuations and scope
Fixed strictly triangular proportions	Branch mean: $\lambda_i \asymp \sqrt{n}$	Branch scale $n^{1/4}$; competing branch weights from (66)
$\delta = d/n^{2/3}$ in a fixed real compact interval	Mean: $u(\delta)n^{2/3} + 2b(\delta)n^{1/3} + O(1)$	Gaussian scale $n^{1/3}$; Theorems 15.1 and 19.1
$c = d/n$ in a fixed positive compact interval	Exact: $d + 2J$	J tends to Poisson(c^{-2}) when c converges; $O(1)$ gap

At the exact boundary $u = 2^{1/3}$. In the negative part of the compact moving window, $U = d + 2J$ holds with probability $1 - O(e^{-\gamma_K n^{2/3}})$, not on every matching. In the exterior it holds identically.

The compact- δ theorem does not permit sending $\delta \rightarrow \pm\infty$ inside its error bound. In particular it does not by itself connect uniformly to the strict-triangle or fixed-exterior theorem through every intermediate imbalance sequence. Such an overlap theorem would require estimates controlling the deterioration of the saddle distance, curvature, and branch comparison as the parameter compact set grows. The regime table is a comparison of proved domains, not an assertion that their error estimates already match.

24 Exact verification and numerical diagnostics

The source archive supplies a standard-library exact core, independent graph enumeration, reproducible reference data, and a deterministic document builder. Its checks are finite verification evidence for algebra and implementation; the analytic all-orders statements are proved in the preceding sections.

24.1 Independent graph recursion

A recursion counts maximal matchings without using factorial branch sums. Its state consists of remaining labeled part counts $r = (r_1, r_2, r_3)$ and an unmatched-part flag i_0 , initially unset. Select a fixed distinguished vertex from the first nonempty part i . Either leave it unmatched, allowed only if the flag is unset or equals i , or pair it with any of the r_j remaining vertices in a different part j , with multiplicity r_j . In the unmatched case set the flag to i ; in a matched case keep it unchanged. The empty state contributes 1. Every matching specifies exactly one decision for the distinguished vertex at each step, and the flag condition is exactly Lemma 13.1, proving the recursion.

The exact core compares this recursion with the full three-branch formula for $0 \leq n \leq 10$, $-n \leq d \leq n + 2$, a total of 143 pairs. These include both parities and $d = -1, 0, 1$. It separately checks the large-branch sum and the boundary prefix. Boundary phase coefficients are computed over $\mathbb{Q}(a)$ with $4a^3 = 1$, with Gaussian moment insertion and relative-to-log conversion. Moving-window formulas are checked at exact rational values of u and at the algebraic boundary specialization; those finite substitutions are coefficient diagnostics, not a substitute for the symbolic rational derivation and analytic proof. Exterior coefficients are independently generated from (119)–(120).

The verifier uses explicit exceptions for failed checks, so optimized Python does not remove them. Guard tests exercise malformed inputs and package invariants. The strict manifest covers every regular payload file and rejects unsafe paths, symlinks, duplicates, unexpected files, and altered bytes. The builder stages into an isolated directory, disables TeX shell escape, and uses deterministic archive metadata. Existing immutable candidate directories are not overwritten. The README gives the exact replay commands and the tool dependencies for PDF rebuilding; the arithmetic verification core needs only Python’s standard library.

[write] In this directory the exact core is `code/178-boundary-code-verify.py` and `code/178-boundary-code-regenerate.py`; the guards and the builder are `code/178-boundary-guard_tests.py`, `code/178-boundary-test_build.py` and `code/178-boundary-build.py`; the optional audits are `code/178-boundary-optional-*.py`, with their outputs in `data/178-boundary-optional-*`; and the guides are `178-boundary-README_CODE.md`, `178-boundary-optional-README.md` and `178-boundary-SOURCE_AUDIT.md`. The manifest checker is shipped once, as Part I’s byte-identical `code/176-balanced-verify_manifest.py`. `Report178.tex`, `Report178.pdf`, the delivered README and `SHA256SUMS.json` are not shipped. The report’s README gives the rerun route.

24.2 Scope of the high-precision calculations

Optional audit scripts and recorded data use arbitrary-precision and symbolic packages separately from the exact standard-library core. Positive terms are summed about the exact integer mode, with geometric omitted-tail bounds from the monotone consecutive-term ratio. Such analytic truncation bounds, evaluated in ordinary high precision, do *not* certify the total rounding error. No outward-rounded interval certificate for the final decimal values is claimed.

For the boundary at $n = 10^7$, successive residuals after lower count corrections have been removed, scaled by N, N^2, N^3, N^4 , respectively, are approximately

$$0.6160568653, \quad 0.1462343995, \quad -0.1574930933, \quad -0.1077401260.$$

Their exact coefficient targets from (105) are approximately

$$0.6153781054, \quad 0.1469654177, \quad -0.1569930080, \quad -0.1076143422.$$

These are convergence diagnostics; equality at finite n is not expected. Recorded relative omitted-tail bounds for those boundary sums are below 4×10^{-82} , interpreted with the numerical qualification above.

For the moving window at $N = 160$, the diagnostic $N \log(Z/Z_{\text{lead}})$ and target $c_1(u)$ are as follows; each target uses its exact $\delta = d/N^2$.

d	Scaled first-log residual	$c_1(u)$
−76799	1.0770450674	1.0749783875
−25599	1.0178141785	1.0173182249
0	0.6151066009	0.6153781054
25601	0.2776274434	0.2775605148
76801	0.0248501879	0.0247405275

Second-log, mean, and variance residuals also approach their targets. The recorded relative omitted-mass bounds are below 4×10^{-72} for these cases. Additional tests include $d = -1, -2, 1$ and exterior ratios near $1/2, 1, 2$. None of the proofs depends on these decimal experiments.

25 Source comparison and open questions

The bounded source audit was conducted on 3 October 2026. It checked the live A297487 entry, primary metadata and accessible papers, a public mathematical report catalogue, and a bounded high-relevance search of previously available reports. It found the balanced Report 176 as the direct predecessor. Its general EGF already implies the exact boundary reduction, while its stated proved asymptotic scope remains balanced. No additional boundary theorem was located in that bounded inspection. Ranked searches, limited result windows, incomplete extraction, and uninspected archives prevent any exhaustive absence claim.

The following distinctions limit the attribution conclusions.

1. Song, Miao, Wang, and Zhao [10] study maximal-matching cardinalities and edge domination in complete multipartite graphs according to the primary abstract. Their full text was not obtained. The title and abstract alone cannot establish or exclude the enumeration refinements here.

2. Azor, Gillis, and Victor [1] give the classical Hermite-pairing framework. Primary metadata and the abstract were checked, but full text was not obtained. The shifted identity (65) is an elementary generating-function consequence within that framework; no priority claim is based on failure to retrieve an earlier formulation.
3. Paris [8] treats unshifted Hermite-product integrals and classical discrete Laplace/Poisson–Jacobi methods. The inspected all-order formula for four equal Hermite indices has a different positive sum and ordinary $1/n$ corrections; the inspected unequal-index end cases are unshifted. Those sections do not directly provide the shifted mixed-index $n^{2/3}$ theorem above. This is a limited theorem comparison, not a survey of all Hermite literature.
4. Hang and Luo [5] give several large-parameter hypergeometric expansions in Section 4, including paired numerator/denominator shifts and specific restrictions on the numbers and integrality of moving parameters. The inspected theorems do not directly cover ${}_2F_2(-n, -n; 1/2, 1; 1/4)$ with two terminating moving numerators and fixed denominators.
5. DLMF Section 16.11 [3] treats large-variable and certain large-parameter cases under parameter restrictions different from the present one. It points to Luke’s Chapter 7 [6], which was not obtained in full. That remaining full-text gap is explicit.

Direct numeric OEIS retrieval was unsuccessful, and other public numeric/graph-string searches did not establish a boundary ID. These failures are not absence certificates. The safe description is a related boundary continuation of A297487 with proved refinements, not a new sequence under that identifier or a globally new asymptotic method. The companion source-audit file records the read-only search scope and useful primary links without making a publication or external-edit claim.

[write] 6 October 2026. The write’s OEIS searches found no entry for the boundary sequence either (front matter). The DLMF entry cited in item 5 has the key `d1mf16` in this report.

Further questions naturally left open are:

1. Establish the actual OEIS identity, if any, of A_n , and resolve the Song, Azor, and Luke full-text comparison gaps.
2. Prove a genuinely uniform overlap theorem as $\delta = d/n^{2/3} \rightarrow \pm\infty$, including intermediate imbalance scales.
3. Make the remainder constants and onsets effective, then derive outward-rounded count and inverse certificates. Ordinary high-precision diagnostics do not accomplish this.
4. Extend the boundary transition to unequal small parts and describe when multiple unmatched-part branches remain comparable near a degenerating triangle inequality.
5. Study sharper tail asymptotics beyond the central Gaussian range. Rare maximum events should continue to use their exact combinatorial numerator rather than a central approximation.

[write] 6 October 2026. These are items 10, 6, 4, 7 and 8 of Section 26. Question 1 stays open: the write’s searches of 6 October 2026 found no OEIS entry for A_n .

26 Further questions and research

This section is added in the write, under Vladimir’s standing rule of 4 October 2026: every claim of the two manuscripts that is not proved here, and every limitation they state, is collected as a question, with its source, the sketch that came with it, and what is missing. The delivered lists (Part I, Section 11.2; Part II, Section 25) are printed in place; a question that the other Part answers is re-scoped here, not dropped. No claim of either manuscript was found wrong.

1. *The parity sector of the balanced count* (Part I, question 1 and Remark 4.2). The expansion (18) is algebraic: its signed Taylor remainder is bounded algebraically, so the exponentially small parity sector of $S_n(v)$ is not identified, and the exact subtraction in (5) does not make the count expansion exponentially accurate. *Missing*: the signed gamma-weighted Poisson expectation $\mathbb{E}[(-1)^P R_n(P) \mathbf{1}_{P \leq n}]$ beyond all algebraic orders.
2. *Marking transition regimes* (Part I, question 2). Theorem 4.1 is uniform only for v in compact subsets of $(0, \infty)$. *Sketch*: the scales $v \rightarrow 0$ and $v \rightarrow \infty$ should interpolate between fixed unmatched counts and macroscopic deficiency. *Missing*: any uniform statement there.
3. *Complex marking and sharper local laws* (Part I, question 3; Part I, Section 6). (31) is a real-parameter statement, with no uniform complex neighbourhood of $v = 1$; Part II proves no complex-marking statement either; its local and total-variation bounds (Theorems 19.1 and 22.1) are of first order. *Missing*: a controlled complex- v expansion and Edgeworth corrections with explicit lattice terms, in either setting.
4. *Effective constants, onsets and certified thresholds* (Part I, question 4; Part II, question 3). All error constants and onsets of both Parts are existential; the threshold brackets (Theorems 9.2 and 21.2) are exact implications, not numerical certificates, and neither Part proves a bare rounding rule. *Sketch*: quantify the Stirling, Chernoff, Taylor and lattice remainders. *Missing*: explicit constants, after which the brackets become executable for every y beyond an explicit bound, with exact counting below it; and outward-rounded count and inverse certificates (Part II says that ordinary high-precision diagnostics do not accomplish this).
5. *Unequal parts and more parts* (Part I, question 5). *Re-scoped* (front matter, “Part II and the question of Part I”): Part II settles the instance $K_{2n+d,n,n}$ across the face $a = b + c$ of the triangle inequality, in the window $d/n^{2/3}$ compact and in the exterior d/n compact in $(0, \infty)$, and gives the competing branch contributions for compact strictly triangular proportions to leading order. Open: (a) a growing or larger fixed number of parts, where the maximality lemma still confines the unmatched vertices to one part but the edge populations are no longer determined by the unmatched count; (b) the strict triangle beyond leading order (Theorem 14.1 has relative error $O(n^{-1/2})$) and Part I’s all-order method uses the symmetry of the balanced case, as Part I says; (c) items 6 and 7.
6. *A uniform overlap between the regimes* (Part II, question 2 and Section 23). Theorem 15.1 does not permit $\delta = d/n^{2/3} \rightarrow \pm\infty$ inside its error bound, and Part II’s regime table “is a comparison of proved domains, not an assertion that their error estimates already match”. *Sketch* (Part II): control the deterioration of the saddle distance, the curvature and the branch comparison as the parameter compact grows. *Missing*: such estimates, for every intermediate imbalance scale.
7. *Unequal small parts near a degenerating triangle* (Part II, question 4; Section 14). Corollary 14.2 and Theorem 14.1 need a compact strict-triangle region, and Part II’s proof “supplies no

estimate uniform as a triangle inequality degenerates”. *Missing*: the boundary transition for K_{p_1,p_2,p_3} with $p_1 \approx p_2 + p_3$ and $p_2 \neq p_3$, and a description of when several unmatched-part branches stay comparable.

8. *Sharper tails* (Part II, question 5). Part II’s laws are central; its exact rare-event formulas ((110) and the exterior maximum probability) use exact numerators, as do Part I’s (Theorem 8.1). *Missing*: tail asymptotics between the central Gaussian range and those exact rare events, in both Parts.
9. *A complete inverse for the boundary sequence* (Part II, Section 21). Part II gives the Lambert initializer and only the leading displacement $-3ax_0^{2/3}/(2\log(2x_0))$, “not a complete ordered reversion series”, and recommends implicit inversion of (111). Part I’s explicit inverse (Theorem 9.3) has $o(1)$ absolute error; its higher fixed orders are left implicit in the models F_J (“Formal Newton expansion . . . gives further explicit terms when desired”). *Sketch*: the operator series of the transseries volume about the factorial core (front matter) produces the formal terms. *Missing*: an ordered expansion in the mixed scales $x_0^{2/3}$, $x_0^{1/3}$, $\log x_0$ with a proved remainder.
10. *The boundary sequence and the unread literature* (Part II, question 1; Part I, Section 11; Part II, Section 25). $M(2n, n, n)$ has no OEIS entry as far as Part II and the write could see (front matter). Neither Part nor the write obtained the full texts of Song–Miao–Wang–Zhao [10], Azor–Gillis–Victor [1] or Luke’s Chapter 7 [6]; Part I says the abstracts “do not certify theorem-by-theorem nonoverlap”. Part II’s audit found that the large-parameter hypergeometric theorems it inspected (Hang–Luo, DLMF §16.11) do not cover ${}_2F_2(-n, -n; \frac{1}{2}, 1; \frac{1}{4})$. *Missing*: the full-text comparisons, and a search beyond the bounded searches of 3 October 2026.
11. *Uncertified numerics* (Part II, Section 24; Part I, Section 10). Part II’s residuals at $n = 10^7$, its window table at $N = 160$ and its omitted-tail bounds are ordinary high-precision diagnostics, “not outward-rounded”; Part I’s numerical reference was “read without executing code”. No theorem rests on them. *Missing*: interval certificates, if wanted (item 4).
12. *Literature and priority*. Both Parts made bounded searches on 3 October 2026 and make no worldwide priority claim. Part I’s leading equivalent is Kotěšovec’s and its positive sum is Howroyd’s (credited); Paris already has the all-order expansion for the nearby four-equal-Hermite (perfect-matching) problem; Part II’s exact identities are presented as classical consequences. *Missing*: as in item 10.

[write] **6 October 2026**. *Answered, re-scoped and checked in this report*. Part I’s question 5 is re-scoped: Part II settles one instance (item 5). Part I’s sentence that the b-file was not retrieved is superseded by the write’s comparison of all 100 terms (note in Section 2; sentence kept). No mathematical statement of the two manuscripts was found wrong, and no published result is corrected.

References

- [1] R. Azor, J. Gillis, and J. D. Victor, Combinatorial applications of Hermite polynomials, *SIAM Journal on Mathematical Analysis* **13** (5) (1982), 879–890. <https://doi.org/10.1137/0513062>. Primary abstract and metadata consulted; full text not obtained. (Part I.) Part II: *Combinatorial Applications of Hermite Polynomials*, *SIAM Journal on Mathematical Analysis* **13** (1982), 879–890. [doi:10.1137/0513062](https://doi.org/10.1137/0513062). Abstract and metadata inspected; full text not obtained.

- [2] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, §5.11, Asymptotic expansions of the gamma function, and §24.2, Bernoulli polynomials. <https://dlmf.nist.gov/5.11>; <https://dlmf.nist.gov/24.2>. (Part I.)
- [3] NIST Digital Library of Mathematical Functions, [Section 16.11, Asymptotic expansions of generalized hypergeometric functions](#). Consulted 3 October 2026. (Part II, where the key was `dlmf`; renamed in the write.)
- [4] M. Drmota, L. Ramos, C. Requilé, and J. Rué, Maximal independent sets and maximal matchings in series-parallel and related graph classes, *29th International Conference on Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms*, LIPIcs **110** (2018), 18:1–18:15. <https://doi.org/10.4230/LIPIcs.AofA.2018.18>. (Part I.)
- [5] P.-C. Hang, M.-J. Luo, *Asymptotics of the Humbert Function Ψ_1 for Two Large Arguments*, SIGMA **20** (2024), 074. [Primary journal PDF](#). (Part II.)
- [6] Y. L. Luke, *The Special Functions and Their Approximations*, Vol. I, Academic Press, 1969, Chapter 7, as referenced by DLMF. Full chapter not obtained for this audit. (Part II.)
- [7] OEIS Foundation Inc., *A297487: Number of maximal matchings in the complete tripartite graph $K_{n,n,n}$* . Entry consulted 3 October 2026. Positive-sum program by Andrew Howroyd (30 December 2017); leading asymptotic and recurrence credited to Václav Kotěšovec (6 February 2026). <https://oeis.org/A297487>. (Part I.) Part II: OEIS Foundation, [A297487, Number of maximal matchings in the complete tripartite graph \$K_{n,n,n}\$](#) . Consulted 3 October 2026. Exact program by A. Howroyd; leading balanced asymptotic credited to V. Kotěšovec. This is not the boundary sequence identifier. Read again at the write, 6 October 2026 (revision #18 of 6 February 2026), with its b-file <https://oeis.org/A297487/b297487.txt> ($n = 1, \dots, 100$, Andrew Howroyd).
- [8] R. B. Paris, Asymptotics of integrals of Hermite polynomials, *Applied Mathematical Sciences* **4** (61) (2010), 3043–3056. <https://www.m-hikari.com/ams/ams-2010/ams-61-64-2010/parisAMS61-64-2010.pdf>. (Part I, which read the full paper; Part II read Section 5 and the unequal-index end cases.)
- [9] Research report 176, *Maximal Matchings in Balanced Complete Tripartite Graphs: All fixed order asymptotics, unmatched vertices, and inverse thresholds*, 3 October 2026. Related predecessor for $K_{n,n,n}$ and the general marked EGF. (Part II.) **[write]** Printed in this report as Part I.
- [10] W. Song, L. Miao, H. Wang, and Y. Zhao, Maximal matching and edge domination in complete multipartite graphs, *International Journal of Computer Mathematics* **91** (5) (2014), 857–862. <https://doi.org/10.1080/00207160.2013.818668>. Primary abstract consulted; full text not obtained. (Part I.) Part II: [doi:10.1080/00207160.2013.818668](https://doi.org/10.1080/00207160.2013.818668). Abstract and metadata inspected; full text not obtained.
- [11] ProveIt, *Transseries and inversion*, the repository’s transseries volume, [Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex](#); Part 0: `p0:prop:factorial-core`, `p0:prop:operator-form`, `p0:thm:staircase`. Added in the write.